SCHOOL OF COMPUTATION,
INFORMATION AND TECHNOLOGY —
TUM SCHOOL OF COMPUTATION,
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Master's Thesis

Uncertainty Quantification for Graph Neural Network Surrogates of Agent-Based Transport Models

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Unsicherheitsquantifizierung für Graph-Neuronale-Netzwerk-Surrogate von agentenbasierten Transportmodellen

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Submission Date: April 15, 2026

I confirm that this master's thesis is my own work and I have documented all sources and material used.

Munich, April 15, 2026 Mohd Zamin Quadri

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Abstract

Agent-based transport simulations are the de facto methodology for evaluating urban policy interventions, enabling planners to assess the impact of infrastructure changes on traffic patterns across entire metropolitan areas. However, the computational burden of such simulations—often requiring hours per scenario—severely limits the scope of policy exploration. Graph Neural Networks (GNNs) have been shown to be effective surrogates for these simulations, offering scenario evaluation times reduced from hours to seconds while maintaining reasonable predictive accuracy.

While the predictive performance of GNN surrogates has been studied, their reliability in deployment settings remains underexplored. When a surrogate encounters scenarios that deviate from its training distribution, predictions may become unreliable without any indication to the end user. This limitation poses significant risks for decision support applications where understanding prediction confidence is as critical as the predictions themselves.

In this thesis, we advance the study of uncertainty quantification (UQ) for GNN-based traffic prediction surrogates. We train and evaluate eight model configurations on a Paris road network dataset comprising 10,000 MATSim scenarios; all reported results are based on a 1,000-scenario (10%) subset unless stated otherwise. The best-performing configuration (Trial 8) achieves $R^2 = 0.5957$, MAE = 3.96 vehicles/hour, and RMSE = 7.12 vehicles/hour on the test set of 100 graphs (31,635 road segments each), yielding 3,163,500 node-level predictions.

We systematically compare three UQ approaches. For Trial 8 on its full 100-graph test set, Monte Carlo (MC) Dropout with $S = 30$ stochastic forward passes achieves Spearman uncertainty–error correlation $\rho = 0.4820$. In a separate ensemble comparison experiment (Experiment A, evaluated on a different subset with known distribution mismatch), MC Dropout achieves $\rho = 0.1600$ versus ensemble variance $\rho = 0.1035$—a 55% relative improvement within that context at substantially lower computational cost. Conformal prediction at the 95% level produces well-calibrated coverage (95.01%) with a quantile of $q = 14.68$ vehicles/hour (primary 50/50 calibration/evaluation split; a 20/80 audit split corroborates at $q = 14.77$), while naïve Gaussian intervals using $k = 1.96\sigma$ achieve only 55.6% empirical coverage—demonstrating that the MC Dropout standard deviation is not a calibrated uncertainty estimate ($k_{95} = 11.34$) without post-hoc correction.

We further demonstrate the practical value of uncertainty-aware prediction: selectively withholding the 50% most uncertain predictions reduces MAE by 41.6% (from 3.94 MC Dropout mean to approximately 2.31 vehicles/hour). A cross-trial replication on Trial 7 confirms that these findings are not specific to the best model: the same qualitative con-

clusions hold across both trials, with conformal prediction restoring near-exact nominal coverage in both cases. We document a significant negative result regarding multi-model ensembling: ensemble experiments with near-identical architectures yield near-zero predictive R^2, a result attributable to data distribution mismatch between ensemble training subsets rather than a fundamental failure of ensembling as a technique.

Our findings establish MC Dropout combined with conformal calibration as a practical, computationally efficient uncertainty quantification strategy for GNN-based transport surrogates, with direct applicability to uncertainty-aware urban planning decision support.

Keywords: Graph Neural Networks, Uncertainty Quantification, Monte Carlo Dropout, Conformal Prediction, Agent-Based Models, Traffic Prediction, Bayesian Deep Learning

Zusammenfassung

Agentenbasierte Verkehrssimulationen sind die etablierte Methodik zur Bewertung urbaner Politikinterventionen und ermöglichen es Planern, die Auswirkungen von Infrastrukturänderungen auf Verkehrsmuster in gesamten Metropolregionen zu bewerten. Der hohe Rechenaufwand solcher Simulationen—oft mehrere Stunden pro Szenario—schränkt jedoch den Umfang der Politikexploration erheblich ein. Graph Neural Networks (GNNs) wurden erfolgreich als Surrogatmodelle für diese Simulationen eingesetzt und bieten eine Reduzierung der Szenariobewertungszeiten von Stunden auf Sekunden bei gleichzeitiger Beibehaltung akzeptabler Vorhersagegenauigkeit.

Während die Vorhersageleistung von GNN-Surrogaten untersucht wurde, bleibt ihre Zuverlässigkeit in realen Einsatzszenarien unzureichend erforscht. Wenn ein Surrogatmodell auf Szenarien trifft, die von seiner Trainingsverteilung abweichen, können Vorhersagen unzuverlässig werden, ohne dass der Endnutzer darauf hingewiesen wird. Diese Einschränkung birgt erhebliche Risiken für Entscheidungsunterstützungsanwendungen, bei denen das Verständnis der Vorhersagekonfidenz ebenso entscheidend ist wie die Vorhersagen selbst.

In dieser Arbeit untersuchen wir systematisch Methoden zur Unsicherheitsquantifizierung (UQ) für GNN-basierte Verkehrsvorhersage-Surrogate. Wir trainieren und evaluieren acht Modellkonfigurationen auf einem Pariser Straßennetz-Datensatz mit 10.000 MATSim-Szenarien; alle berichteten Ergebnisse basieren auf einer 1.000-Szenario-Teilmenge (10%), sofern nicht anders angegeben. Die beste Konfiguration (Trial 8) erreicht $R^2 = 0{,}5957$, MAE $= 3{,}96$ Fahrzeuge/Stunde und RMSE $= 7{,}12$ Fahrzeuge/Stunde auf dem Testset von 100 Graphen (je 31.635 Straßenabschnitte), was 3.163.500 Vorhersagen auf Knotenebene ergibt.

Wir vergleichen drei UQ-Ansätze. Für Trial 8 auf dem vollständigen 100-Graphen-Testset erreicht Monte Carlo (MC) Dropout mit $S = 30$ stochastischen Vorwärtsdurchläufen eine Spearman-Korrelation von $\rho = 0{,}4820$ zwischen Unsicherheit und Fehler. In einem separaten Ensemble-Vergleichsexperiment (Experiment A, ausgewertet auf einer anderen Teilmenge mit bekanntem Verteilungsmismatch) erreicht MC Dropout $\rho = 0{,}1600$ gegenüber Ensemble-Varianz $\rho = 0{,}1035$—eine 55%ige relative Verbesserung innerhalb dieses Kontexts bei deutlich geringerem Rechenaufwand. Konforme Vorhersage auf dem 95%-Niveau liefert gut kalibrierte Abdeckung (95,01%) mit einem Quantil von $q = 14{,}68$ Fahrzeugen/Stunde (primärer 50/50-Kalibrierungs-/Evaluierungssplit; ein 20/80-Auditsplit bestätigt bei $q = 14{,}77$), während naive Gauß'sche Intervalle mit $k = 1{,}96\sigma$ nur eine empirische Abdeckung von 55,6% erreichen—was zeigt, dass die MC-Dropout-Standardabweichung ohne Post-hoc-Korrektur keine kalibrierte Unsicherheitsschätzung darstellt ($k_{95} = 11{,}34$).

Darüber hinaus demonstrieren wir den praktischen Nutzen unsicherheitsbewusster Vorhersage: Das selektive Zurückhalten der 50% unsichersten Vorhersagen reduziert den MAE um 41,6% (von 3,94 auf ca. 2,31 Fahrzeuge/Stunde, gemessen am MC-Dropout-Mittelwert als Ausgangspunkt). Eine Cross-Trial-Replikation auf Trial 7 bestätigt, dass diese Ergebnisse nicht spezifisch für das beste Modell sind: Die gleichen qualitativen Schlussfolgerungen gelten für beide Trials, wobei konforme Vorhersage in beiden Fällen nahezu exakte nominale Abdeckung wiederherstellt. Wir dokumentieren ein signifikantes negatives Ergebnis bezüglich Multi-Modell-Ensembling: Ensemble-Experimente mit nahezu identischen Architekturen liefern einen nahezu null R^2, was auf einen Datensatz-Verteilungsmismatch zwischen den Trainingsteilmengen der Ensembles zurückzuführen ist.

Unsere Ergebnisse etablieren MC Dropout kombiniert mit konformer Kalibrierung als praktische und recheneffiziente UQ-Strategie für GNN-basierte Transportsurrogate mit direkter Anwendbarkeit für unsicherheitsbewusste Entscheidungsunterstützung in der Stadtplanung.

Schlüsselwörter: Graph Neural Networks, Unsicherheitsquantifizierung, Monte Carlo Dropout, Konforme Vorhersage, Agentenbasierte Modelle, Verkehrsvorhersage, Bayesianisches Deep Learning

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1 Introduction

Urban transportation networks are at once deeply familiar and poorly understood: millions of daily journeys produce emergent traffic patterns that resist simple analytical description [HNA16]. When a city government considers closing a major arterial, reducing road capacity to encourage modal shift, or investing in new infrastructure, it faces a fundamental question: what will actually happen to traffic? Agent-based models (ABMs) have become the standard methodology for answering such questions, simulating the decisions of individual travellers to predict how policy interventions reshape flow across the network [Bon02; RG19].

The computational demands of these simulations, however, create a real bottleneck in the policy evaluation process. Simulating a single scenario for a large metropolitan area like Paris can require eight or more hours on modern hardware—severely limiting the number of alternatives that planners can feasibly explore within project timelines [Nat+25].

Machine learning surrogates offer a practical solution to this challenge. By training neural networks to approximate simulation outputs, scenario evaluation times can be reduced from hours to seconds [Wil+22]. Recent work by Natterer et al. [Nat+25] demonstrated that Graph Neural Networks (GNNs) are particularly well-suited for this task: they naturally capture the graph structure of road networks while learning complex relationships between policy interventions and traffic flow changes.

Yet accuracy alone is not sufficient for deployment in decision support systems. When a surrogate model encounters scenarios that differ substantially from its training distribution, its predictions may become unreliable—and without uncertainty quantification (UQ), users have no principled way to identify these problematic cases. This limits the trustworthiness of surrogate-based planning tools.

This thesis addresses that gap directly. We investigate uncertainty quantification methods for GNN-based traffic surrogates, conducting a systematic comparison of Monte Carlo (MC) Dropout against ensemble-based approaches and conformal prediction. The goal is to establish which UQ methods are most effective for graph-structured traffic prediction, and to provide practical guidance for deploying uncertainty-aware surrogates in urban planning. All experiments use 1,000 of 10,000 available MATSim scenarios (10% subset).

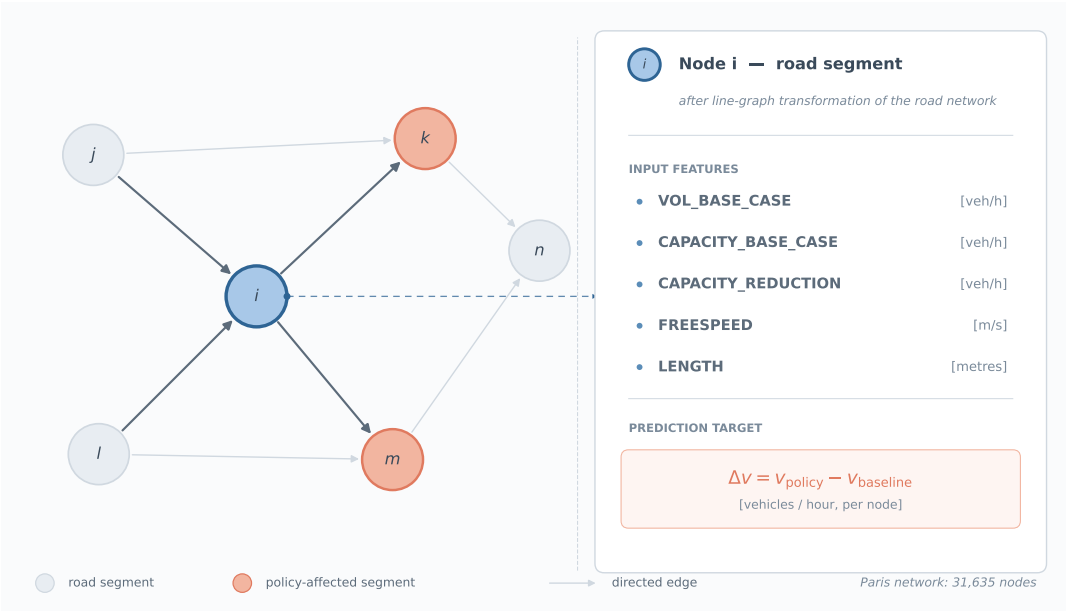

Figure 1.1: Directed line-graph representation of the Paris road network used by the GNN surrogate. Each node represents one road segment with five input features (listed right); coral-shaded nodes carry a non-zero `CAPACITY_REDUCTION` due to the policy intervention. The per-node prediction target is $\Delta v = v_{\text{policy}} - v_{\text{baseline}}$ (veh/h). The Paris network contains 31,635 nodes. Source: Natterer, Rao, Tejada Lapuerta, et al. [Nat+25].

1.1 Problem Statement

The deployment of GNN surrogates for traffic prediction presents a fundamental challenge: models must generalize to policy scenarios that may differ substantially from their training distribution. In classical supervised learning settings, this distribution shift manifests as degraded prediction accuracy. In decision support applications, however, the consequences are more severe—planners may unknowingly base critical infrastructure decisions on unreliable predictions.

The research gap we address comprises three interconnected components:

1. **Epistemic uncertainty in GNN surrogates:** While uncertainty quantification has been extensively studied in computer vision and natural language processing, its application to GNN-based traffic prediction remains largely unexplored. The graph structure of road networks introduces unique challenges, as uncertainty may propagate through connected road segments.

2. **Method comparison:** MC Dropout and deep ensembles represent the two dominant paradigms for uncertainty quantification in deep learning [GG16; LPB17]. However, their comparative effectiveness for graph-structured traffic data has not been systematically evaluated.

3. **Practical deployment guidance:** Urban planners require concrete recommendations about which UQ method to adopt, how to configure its parameters, and how to interpret the resulting uncertainty estimates in decision-making contexts.

1.2 Research Questions

This thesis addresses three closely related research questions:

RQ1: How effectively does MC Dropout capture epistemic uncertainty in GNN-based traffic surrogates, as measured by the correlation between predicted uncertainty and actual prediction error?

RQ2: How does MC Dropout compare to ensemble-based uncertainty quantification—both variance across training runs and multi-model ensembles—in terms of uncertainty quality and computational efficiency?

RQ3: Does combining uncertainty estimates from architecturally identical models provide benefits over single-model UQ methods, and what role does model diversity play in ensemble effectiveness?

1.3 Contributions

The contributions of this thesis are as follows:

1. **Systematic UQ evaluation:** We conduct the first comprehensive comparison of uncertainty quantification methods for GNN-based traffic surrogates, evaluating MC Dropout, ensemble variance, and multi-model ensembles on a large-scale Paris road network dataset with 3.16 million node-level predictions across 100 test scenarios.

2. **Empirical findings on method effectiveness:** For Trial 8 (best model), MC Dropout achieves Spearman $\rho = 0.4820$ between predicted uncertainty and absolute error (100 test graphs, 3,163,500 predictions). In a separate ensemble comparison (Experiment A), MC Dropout achieves $\rho = 0.1600$ vs ensemble variance $\rho = 0.1035$ within a mismatched evaluation subset—a 55% relative improvement in that context.

3. **Negative result on homogeneous ensembles:** Ensemble methods built from architecturally identical models fail to improve prediction accuracy over the single best model, providing empirical evidence that architectural diversity is necessary for effective ensembling.

4. **Deployment recommendations:** We provide concrete guidance for practitioners implementing uncertainty quantification in traffic GNN systems, including parameter choices ($S = 30$ MC samples), model selection strategies, and frameworks for interpreting uncertainty estimates.

1.4 Thesis Structure

The remainder of this thesis is organized as follows:

Chapter 2: Background and Related Work establishes the theoretical foundations of this research. Section 2.1 introduces agent-based transport models with emphasis on the MATSim framework. Section 2.2 presents the fundamentals of graph neural networks and their application to traffic prediction. Section 2.3 reviews uncertainty quantification methods in deep learning, including Bayesian approaches and ensemble methods.

Chapter 3: Methodology describes the technical approach adopted in this thesis. We present the PointNetTransfGAT architecture, detail our implementations of MC Dropout and ensemble-based UQ methods, and define the evaluation metrics used to assess uncertainty quality.

Chapter 4: Experimental Setup presents the Paris road network dataset, training protocol, and experimental design. We describe two complementary experiments: Experiment A comparing MC Dropout against ensemble variance on a single model, and Experiment B evaluating multi-model ensemble approaches.

Chapter 5: Results reports the findings from both experiments, including quantitative comparisons of UQ methods and analysis of the generalization gap observed between validation and test performance.

Chapter 6: Discussion interprets results in the context of prior work, addresses limitations of our study, and discusses implications for deploying uncertainty-aware traffic prediction systems.

Chapter 7: Conclusion summarizes the key findings, revisits the research questions, and outlines directions for future work.

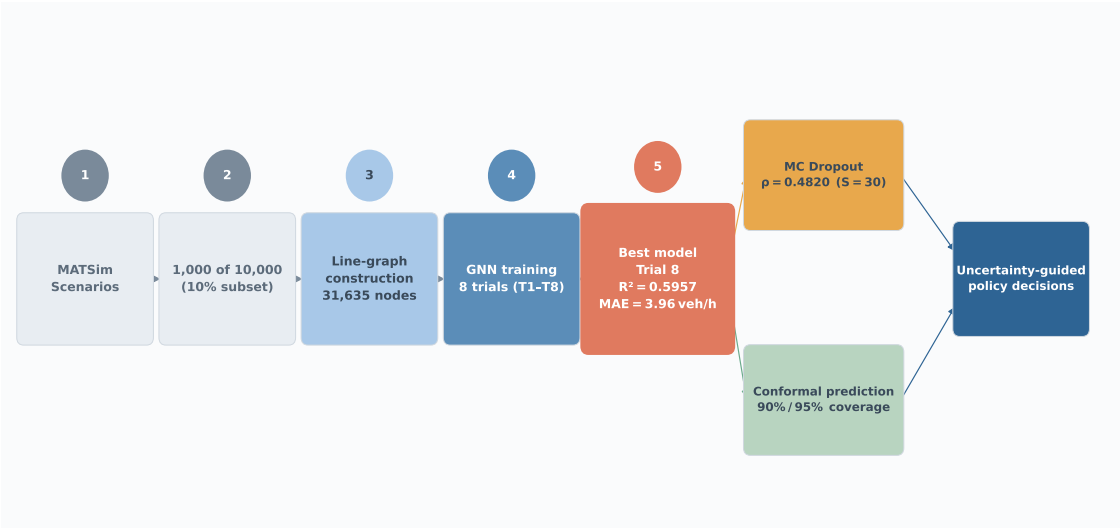

Figure 1.2: Overall thesis pipeline. MATSim agent-based simulations (1,000 of 10,000 available scenarios, 10% subset) are converted to line-graph representations and used to train eight GNN trials. The best model (T8, $R^2 = 0.5957$) is equipped with MC Dropout uncertainty ($\rho = 0.4820$) and conformal prediction intervals (90%/95% coverage guaranteed), feeding into an uncertainty-guided policy decision framework.

2 Background and Related Work

This chapter sets out the theoretical background for the work that follows. Section 2.1 covers agent-based transport models and the MATSim simulation framework. Section 2.2 introduces graph neural networks and their application to traffic prediction. Section 2.3 reviews uncertainty quantification in deep learning, with particular attention to Monte Carlo Dropout, deep ensembles, and conformal prediction. Section 2.4 situates this work within the existing literature on ML surrogates and uncertainty quantification for graph-based predictions.

2.1 Agent-Based Transport Models

Agent-based models (ABMs) simulate complex systems by modelling individual entities and their interactions [Bon02; RG19]. Rather than describing aggregate traffic flow with differential equations, they let emergent patterns arise from the decisions of many autonomous agents. In transport applications, agents typically represent travellers who decide on routes, departure times, and travel modes. The resulting macroscopic traffic patterns emerge from—and can only be understood through—the collective effect of these individual choices.

Definition 2.1 (Agent-Based Model)**.** An agent-based model consists of a set of autonomous agents $\mathcal{A} = \{a_1, a_2, \ldots, a_N\}$, an environment $\mathcal{E}$ in which agents operate, and a set of rules $\mathcal{R}$ governing agent behavior and interactions. Each agent a_i maintains an internal state s_i and makes decisions based on local observations and objectives.

2.1.1 MATSim Framework

The Multi-Agent Transport Simulation (MATSim) framework [HNA16] is a widely-used open-source platform for large-scale transport simulations. Each synthetic agent has a daily activity plan specifying where they need to go and when; the simulation handles activity scheduling and location choice, mode selection across car, public transit, cycling, and walking, route choice based on experienced travel times, and dynamic traffic assignment through queue-based network loading. A typical run iterates through multiple synthetic "days," with agents learning from experience and gradually revising their plans. This evolutionary process converges toward a user equilibrium where no agent can unilaterally improve their travel time [HNA16].

2.1.2 Computational Challenges

While MATSim provides high-fidelity traffic predictions, the computational cost is substantial. Simulating the Paris metropolitan region—our study area—requires more than eight hours per scenario on modern hardware, significant memory for agent state tracking, and multiple iterations to reach equilibrium. These constraints motivate the development of surrogate models that can approximate MATSim outputs at a fraction of the computational cost.

2.2 Graph Neural Networks

Graph Neural Networks (GNNs) extend deep learning to graph-structured data, making them natural candidates for traffic network analysis [Sca+09; KW17; Wu+20]. The formal definition of graphs and the message passing framework underlying modern GNNs are introduced below.

Definition 2.2 (Graph)**.** A graph is a tuple $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{X}, \mathbf{E})$ defined by a nonempty finite set of nodes $\mathcal{V}$, a set of edges $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$, node features $\mathbf{X} \in \mathbb{R}^{|\mathcal{V}| \times d_n}$, and edge features $\mathbf{E} \in \mathbb{R}^{|\mathcal{E}| \times d_e}$.

The neighbourhood of a node v, denoted $\mathcal{N}(v)$, is the set of all nodes connected to v by an edge: $\mathcal{N}(v) = \{u \in \mathcal{V} \mid (u, v) \in \mathcal{E}\}$.

2.2.1 Message Passing Framework

Modern GNNs follow the message passing paradigm [Gil+17], where node representations are iteratively updated through neighbourhood aggregation:

Definition 2.3 (Message Passing Neural Network)**.** Given initial node representations $\mathbf{h}_v^{(0)} = \mathbf{x}_v$, a message passing neural network updates node representations over K layers through the following operations:

$$\mathbf{m}_v^{(k)} = \mathrm{A}^{(k)} \left(\left\{ \mathbf{h}_u^{(k-1)} : u \in \mathcal{N}(v) \right\} \right) \tag{2.1}$$

$$\mathbf{h}_v^{(k)} = \mathrm{U}^{(k)} \left(\mathbf{h}_v^{(k-1)}, \mathbf{m}_v^{(k)} \right) \tag{2.2}$$

where $\mathbf{h}_v^{(k)}$ denotes the representation of node v at layer k, $\mathrm{A}^{(k)}$ is a permutation-invariant function, and $\mathrm{U}^{(k)}$ combines the aggregated neighbourhood information with the node's previous representation.

2.2.2 Graph Attention Networks

Graph Attention Networks (GATs) [Vel+18] introduce attention mechanisms to weight neighbor contributions dynamically:

$$\alpha_{ij} = \frac{\exp\left(\text{LeakyReLU}\left(\mathbf{a}^T[\mathbf{W}\mathbf{h}_i \| \mathbf{W}\mathbf{h}_j]\right)\right)}{\sum_{k\in\mathcal{N}(i)} \exp\left(\text{LeakyReLU}\left(\mathbf{a}^T[\mathbf{W}\mathbf{h}_i \| \mathbf{W}\mathbf{h}_k]\right)\right)} \tag{2.3}$$

where α_{ij} represents the attention coefficient between nodes i and j, $\mathbf{W}$ is a learnable weight matrix, and $\mathbf{a}$ is the attention vector. This attention mechanism enables the model to learn which neighbors are most relevant for each prediction.

2.2.3 GNNs for Traffic Prediction

Recent work has demonstrated strong GNN performance on traffic prediction tasks [Li+18; YYZ18; ZSZ+20; JL22]. The dominant approach combines spatial message-passing over the road graph with temporal sequence modelling using recurrent or attention-based layers. Li et al. [Li+18] introduced diffusion-based graph convolutions to capture directional traffic flow, while Yu et al. [YYZ18] combined graph convolutions with gated recurrent units for short-term speed forecasting. Subsequent work extended these ideas to multi-hop attention and multi-scale spatial representations [JL22]. A common thread is the treatment of intersections as graph nodes, with edges representing road links between them.

An important representational choice, adopted in this thesis following Natterer, Rao, Tejada Lapuerta, et al. [Nat+25], is to represent road segments—rather than intersections—as graph nodes. This *line graph* transformation assigns each road link its own node, enabling the model to predict link-level traffic changes directly. A consequence is that each node must carry the geometric coordinates of its corresponding road segment; in the PointNetTransfGAT architecture, this is handled by providing the 2D start and end positions of each segment as auxiliary positional inputs to the PointNetConv layers, as described in Chapter 3.

2.3 Uncertainty Quantification in Deep Learning

Uncertainty quantification (UQ) enables models to express confidence in their predictions [APH+21; GT+23]. Understanding uncertainty is critical for deploying machine learning models in decision support applications where reliability matters.

Definition 2.4 (Aleatoric Uncertainty)**.** Aleatoric uncertainty, also known as data uncertainty, arises from inherent randomness in the observations. This uncertainty is irreducible and cannot be eliminated by collecting more data. In traffic prediction, day-to-day variation in travel demand represents aleatoric uncertainty.

Definition 2.5 (Epistemic Uncertainty)**.** Epistemic uncertainty, also known as model uncertainty, arises from limited training data or model capacity. This uncertainty is reducible and decreases as more data becomes available. Epistemic uncertainty is the primary focus of this thesis.

2.3.1 Bayesian Neural Networks

Bayesian neural networks (BNNs) place distributions over network weights rather than point estimates [Mac92; Nea96]. For a network with weights $\mathbf{w}$, we seek the posterior distribution:

$$p(\mathbf{w}|\mathcal{D}) = \frac{p(\mathcal{D}|\mathbf{w})p(\mathbf{w})}{p(\mathcal{D})} \tag{2.4}$$

where $\mathcal{D}$ denotes the training data, $p(\mathcal{D}|\mathbf{w})$ is the likelihood, and $p(\mathbf{w})$ is the prior over weights. Exact posterior inference is intractable for neural networks due to the high dimensionality of the weight space, motivating the development of approximate inference methods.

2.3.2 Monte Carlo Dropout

Gal and Ghahramani [GG16] established a theoretical connection between dropout and approximate Bayesian inference. By keeping dropout active at test time and performing T forward passes, we obtain samples from an approximate posterior distribution:

Definition 2.6 (MC Dropout Uncertainty Estimation)**.** For an input $\mathbf{x}$, MC Dropout performs T stochastic forward passes with dropout enabled:

$$\hat{y}_t = f_{\hat{\mathbf{w}}_t}(\mathbf{x}), \quad t = 1, \ldots, T \tag{2.5}$$

where $\hat{\mathbf{w}}_t$ represents the weights after applying dropout mask t. The predictive mean and variance are estimated as:

$$\mathbb{E}[y] \approx \frac{1}{T}\sum_{t=1}^{T} \hat{y}_t \tag{2.6}$$

$$\mathrm{Var}[y] \approx \frac{1}{T}\sum_{t=1}^{T} (\hat{y}_t - \mathbb{E}[y])^2 \tag{2.7}$$

MC Dropout is particularly attractive for practical applications because it requires no architectural modifications—only keeping dropout active during inference. This makes it straightforward to add uncertainty quantification to existing trained models. In our experiments (Chapter 3), we use $S = 30$ stochastic forward passes (denoted S to distinguish from the general T above).

2.3.3 Deep Ensembles

Deep ensembles [LPB17] take an alternative approach by training M networks independently with different random initializations:

Definition 2.7 (Deep Ensemble Uncertainty Estimation)**.** Given M independently trained networks with weights $\{\mathbf{w}_1, \ldots, \mathbf{w}_M\}$, the ensemble prediction and uncertainty are:

$$\mathbb{E}[y] = \frac{1}{M} \sum_{m=1}^{M} f_{\mathbf{w}_m}(\mathbf{x}) \tag{2.8}$$

$$\mathrm{Var}[y] = \frac{1}{M} \sum_{m=1}^{M} \left(f_{\mathbf{w}_m}(\mathbf{x}) - \mathbb{E}[y]\right)^2 \tag{2.9}$$

Ensembles often provide well-calibrated uncertainty estimates but require training and maintaining multiple models, increasing computational cost by a factor of M. This trade-off between uncertainty quality and computational efficiency is a central consideration in our experimental evaluation.

2.3.4 Conformal Prediction

Conformal prediction [AB23; VGS05] is a distribution-free framework for constructing prediction intervals with guaranteed finite-sample coverage. Unlike Bayesian methods, conformal prediction requires no assumptions about the form of the model's predictive distribution—only *exchangeability* of the data.

Definition 2.8 (Split Conformal Prediction)**.** Given a pre-trained model f, a calibration set $\mathcal{D}_{\text{cal}} = \{(x_i, y_i)\}_{i=1}^{n}$, and a desired coverage level $1 - \alpha$, split conformal prediction proceeds as follows:

1. Compute nonconformity scores $s_i = |y_i - f(x_i)|$ for each calibration point.
2. Set the quantile threshold $q = \text{Quantile}\left(s_1, \ldots, s_n;\ \frac{\lceil (n+1)(1-\alpha) \rceil}{n}\right)$.
3. For a new test point x_{new}, construct the prediction interval $\mathcal{C}(x_{\text{new}}) = [f(x_{\text{new}}) - q,\ f(x_{\text{new}}) + q]$.

The key property is a finite-sample marginal coverage guarantee: for any new test point drawn exchangeably from the same distribution,

$$\Pr(y_{\text{new}} \in \mathcal{C}(x_{\text{new}})) \geq 1 - \alpha. \tag{2.10}$$

This guarantee is *model-agnostic* and holds regardless of whether the underlying model's uncertainty estimates are well-calibrated. The trade-off is that conformal intervals are fixed-width (not adaptive to individual inputs), so the coverage guarantee is marginal rather than conditional on each input.

In our experiments (Section 5.5), we apply split conformal prediction to Trial 8 using the absolute residual as the nonconformity score, with a 50/50 calibration–evaluation split of the 100 test graphs.

2.3.5 Evaluating Uncertainty Quality

High-quality uncertainty estimates should correlate with actual prediction errors: the model should express higher uncertainty when it is more likely to be wrong. Three metrics are used to assess this in our experiments. The **Spearman rank correlation** (ρ) measures the monotonic relationship between predicted uncertainty and absolute prediction error; values range from -1 to $+1$, with positive values indicating that higher uncertainty corresponds to higher error. **Calibration curves** assess whether $p\%$ prediction intervals actually contain $p\%$ of true values, with well-calibrated models tracking the diagonal. Finally, the **negative log-likelihood (NLL)** is a proper scoring rule that jointly evaluates prediction accuracy and calibration in a single quantity.

2.4 Related Work

2.4.1 Machine Learning Surrogates for Transport Models

Machine learning surrogates for transport simulations have received increasing attention in recent years [Nat+25]. Historically, surrogate approaches in the transport domain relied on statistical emulation or shallow regression models trained to reproduce aggregate outputs of expensive simulation runs. Willard et al. [Wil+22] provide a broad review of physics-informed machine learning methods used as surrogates across scientific domains, highlighting the tension between data efficiency and physical consistency—a tension that persists when replacing agent-based simulations with learned approximations. More recently, graph-based representations have proven particularly effective for transport surrogates because they preserve the network topology essential for capturing traffic propagation effects [Nat+25].

The PointNetTransfGAT architecture [Nat+25] combines ideas from PointNet, transformers, and graph attention networks to achieve strong performance on traffic prediction from policy scenarios. This architecture forms the foundation for the uncertainty quantification experiments in this thesis.

2.4.2 Uncertainty Quantification for Traffic Prediction

Uncertainty quantification in traffic prediction remains relatively underexplored. Wang et al. [Wan+23] apply probabilistic graph neural networks to spatiotemporal travel demand forecasting — a temporal task in which the model ingests a sequence of observed demand snapshots and predicts future values. This differs fundamentally from the setting studied here: predicting the *change* in link-level traffic volume induced by a transport policy scenario is a static, cross-sectional regression problem with no temporal component. The distributional challenges and appropriate UQ methods therefore differ substantially. To our knowledge, the combination of GNN-based surrogates with distribution-free coverage guarantees via conformal prediction has not been previously explored in the transport literature, and this work addresses that gap.

2.4.3 Uncertainty Quantification for GNNs

Recent work has extended UQ methods to GNNs [Zha+19; Has+20], though primarily for node classification tasks rather than regression. Bayesian GNNs [Zha+19] place prior distributions over graph convolutional weights and perform approximate posterior inference, while DropEdge-style approaches [Has+20] use stochastic edge dropout as a form of variational inference over graph structure. Key findings suggest that graph structure can both help (by providing regularization through neighbourhood aggregation) and hurt (by propagating uncertainty through connected nodes) prediction confidence. The node-level regression setting studied here — where each node represents a road segment and the prediction target is a continuous traffic flow change — introduces different distributional challenges than the categorical outputs most commonly studied in this literature. The interplay between graph structure and uncertainty quantification in regression settings therefore remains an open research question that this thesis helps to address.

3 Methodology

This chapter describes the GNN architecture adopted for traffic prediction, the three uncertainty quantification methods evaluated in this thesis, and the metrics used to assess both predictive accuracy and uncertainty quality.

3.1 Problem Formulation

We formulate traffic prediction as a node-level regression task on a directed graph. Given a road network $G = (V, E)$ where nodes represent road segments and edges represent connectivity, we aim to predict the change in traffic volume Δv_i for each node i after a policy intervention.

3.1.1 Input Features

Each node i is characterized by five input features:

Table 3.1: Input features for each road segment (node).

Feature	Unit	Description
`VOL_BASE_CASE`	vehicles/hour	Traffic volume in baseline scenario
`CAPACITY_BASE_CASE`	vehicles/hour	Road segment capacity
`CAPACITY_REDUCTION`	vehicles/hour	Policy-induced capacity reduction (absolute, ≤ 0)
`FREESPEED`	m/s	Free-flow speed limit
`LENGTH`	meters	Road segment length

3.1.2 Target Variable

The target variable is:

$$\Delta v_i = v_i^{\text{policy}} - v_i^{\text{baseline}} \tag{3.1}$$

representing the change in traffic volume on segment i after the policy intervention is applied.

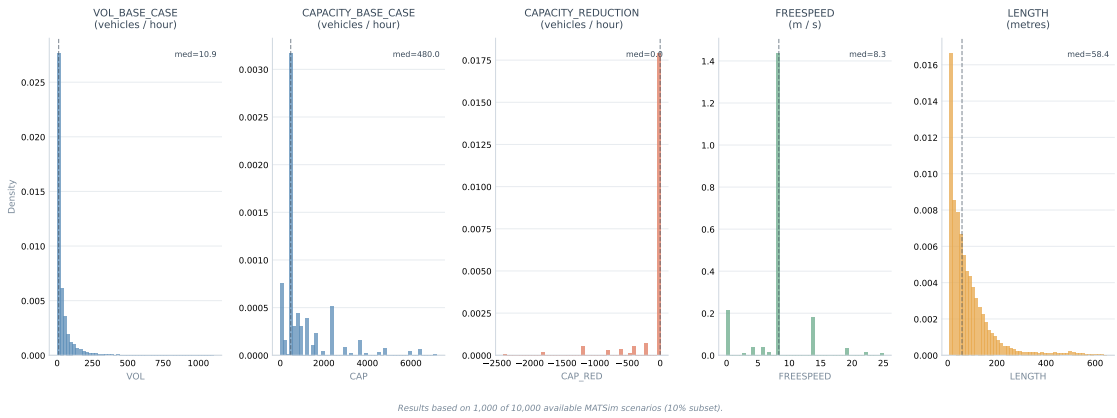

Figure 3.1: Distribution of the five input features across 200 training graphs (6,327,000 node observations). Dashed vertical lines mark the median. `VOL_BASE_CASE` and `CAPACITY_BASE_CASE` are right-skewed, reflecting the heavy-tailed nature of urban traffic volumes; `FREESPEED` is concentrated at a few discrete speed-limit values; `LENGTH` spans several orders of magnitude. *Results based on 1,000 of 10,000 available MATSim scenarios (10% subset).*

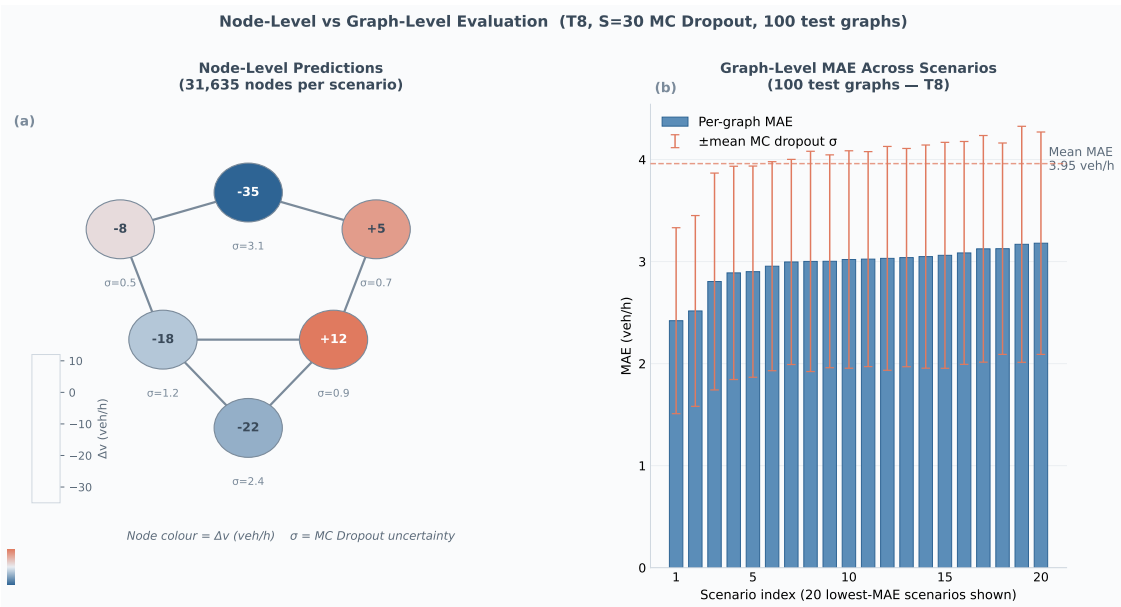

Figure 3.2: Node-level predictions (Δv per node, with per-node MC Dropout σ) and graph-level evaluation (MAE per scenario with uncertainty error bars). T8: 100 test graphs $\times$ 31,635 nodes $=$ 3,163,500 total predictions. Source: `mc_dropout_full_metrics_model8_mc30_100graphs.json`.

3.2 Model Architecture

We use the PointNetTransfGAT architecture developed by Natterer et al. [Nat+25], which combines ideas from PointNet [Qi+17], graph transformers [Shi+21], and graph attention networks [Vel+18] into a single end-to-end model for node-level regression on road networks.

Note on architectural variants: Trial 1 uses a `Linear(64→1)` final prediction layer, while Trials 2–8 use a `GATConv(64→1)` final layer. This difference makes Trial 1 results not directly comparable with Trials 2–8, and Trial 1 is excluded from MC Dropout UQ experiments (effective dropout = 0.0).

3.2.1 Geometric Representation

A key design choice in the architecture is how it encodes the physical geometry of road segments. Each node carries two 2D positional coordinates: the start point of its corresponding road segment, $\mathbf{pos}[:, 0, :] \in \mathbb{R}^{N\times 2}$, and its end point, $\mathbf{pos}[:, 1, :] \in \mathbb{R}^{N\times 2}$. The two PointNetConv layers [Qi+17] exploit these coordinates by computing, for each node i and neighbour j, the relative displacement $\mathbf{pos}_j - \mathbf{pos}_i$ as part of the local MLP input. This means the first PointNetConv uses the start-point geometry of each segment relative to its neighbours, while the second uses the end-point geometry. The effect is that the model learns not only the graph-topological neighbourhood of each segment but also the spatial proximity of where segments begin and end in the 2D plane—an inductive bias well-suited to the geometry of real road networks.

3.2.2 Layer-by-Layer Architecture

Table 3.2 details the complete forward pass through the network. All dimensions and layer specifications are verified directly from `scripts/gnn/models/point_net_transf_gat.py`.

In Stage 1, the local MLP receives a 7-dimensional input: the 5 node features concatenated with the 2D start-position displacement $(\mathbf{pos}_j - \mathbf{pos}_i)[:, 0, :]$. In Stage 2, the local MLP receives 514 dimensions: the 512-dimensional embedding from Stage 1 concatenated with the 2D end-position displacement. The TransformerConv layers [Shi+21] apply multi-head attention over the graph edges; the GATConv layers [Vel+18] aggregate neighbourhood information with learned attention weights.

Dropout placement: Dropout is applied within the local and global MLPs of both PointNetConv layers, and between the TransformerConv layers. All dropout calls are gated by the `use_dropout` flag (default: `False`); the flag must be set to `True` at inference time to activate MC Dropout uncertainty estimation. The architecture also includes a `predict_mode_stats` branch (a TransformerEncoder-based mode predictor) that was implemented in code but was not activated in any of the eight training trials reported in this thesis.

Table 3.2: PointNetTransfGAT layer progression. Input: node features $\mathbf{x} \in \mathbb{R}^{N\times 5}$, positional coordinates $\mathbf{pos} \in \mathbb{R}^{N\times 3\times 2}$ (start, end, and midpoint coordinates; only start and end are used in the forward pass). All dimensions verified from `point_net_transf_gat.py`.

Stage	Layer	Operation	In dim	Out dim
1	PointNetConv-1	Local MLP: 7 → 256 (feat+pos_start); global MLP: 256 → 512 → 512	5	512
2	PointNetConv-2	Local MLP: 514 → 256 (feat+pos_end); global MLP: 256 → 512 → 128	512	128
3	TransformerConv-1	4 attention heads, 64 dims/head	128	256
4	TransformerConv-2	4 attention heads, 128 dims/head	256	512
5	GATConv	1 head	512	64
6a	GATConv (final)	1 head; **Trials 2–8**	64	1
6b	Linear (final)	Point projection; **Trial 1 only**	64	1

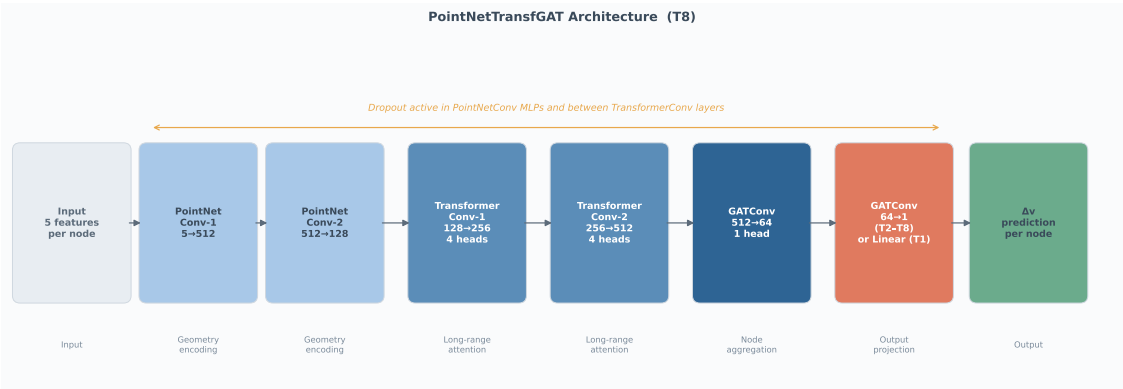

Figure 3.3: PointNetTransfGAT architecture. Two PointNetConv layers encode start- and end-point geometry (512 and 128 channels); two TransformerConv layers capture long-range attention (256 and 512 channels, 4 heads each); two GATConv layers produce the final embedding and scalar prediction (64 then 1 channel). Trial 1 replaces the final `GATConv(64→1)` with `Linear(64→1)`. Verified from `point_net_transf_gat.py`.

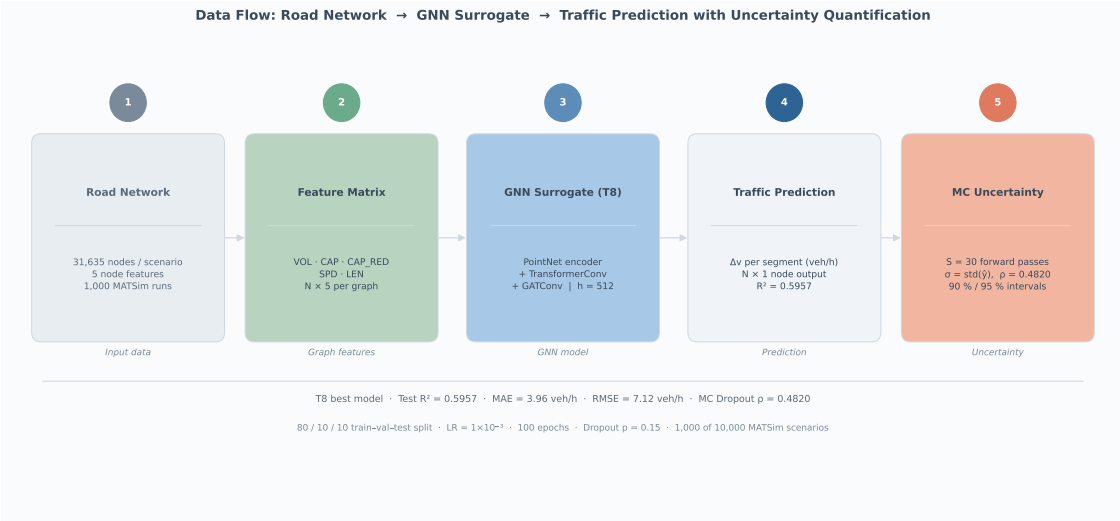

Figure 3.4: End-to-end data flow from road network to traffic prediction with uncertainty. The Paris road network (31,635 nodes) is encoded as a feature matrix ($N \times 5$) with positional coordinates, processed through the GNN, and produces per-node Δv predictions with MC Dropout uncertainty σ. T8 achieves Spearman $\rho = 0.4820$ (100 test graphs, $S = 30$).

3.2.3 Hyperparameter Summary

3.3 Uncertainty Quantification Methods

Three UQ approaches are implemented and compared in this thesis.

3.3.1 Method 1: MC Dropout

MC Dropout [GG16; Gal16] enables uncertainty estimation by keeping dropout active during inference:

We use $S = 30$ MC samples, balancing computational cost against estimate stability.

3.3.2 Method 2: Deep Ensembles

Ensemble variance measures disagreement across independent training runs of the same model:

$$\sigma_{\text{ens}} = \sqrt{\frac{1}{M} \sum_{m=1}^{M} (f_m(\mathbf{x}) - \bar{f}(\mathbf{x}))^2} \tag{3.2}$$

where $M = 5$ is the number of independently trained models.

Table 3.3: Key hyperparameters of the PointNetTransfGAT model, grouped by trial. Full per-trial configurations are given in Table 4.2.

Parameter	Value / Range	Trials
Hidden channels (PointNet local, GAT)	64	All
Hidden channels (TransformerConv)	256 / 512	All
TransformerConv attention heads	4 (both layers)	All
GATConv attention heads	1	All
Dropout rate	0.2	T8
	0.3	T5–T7
	0.0	T1, T3
Learning rate	3×10^{-4} to 10^{-3}	varies
Batch size	8	T5–T8
	16	T2–T4
	32	T1

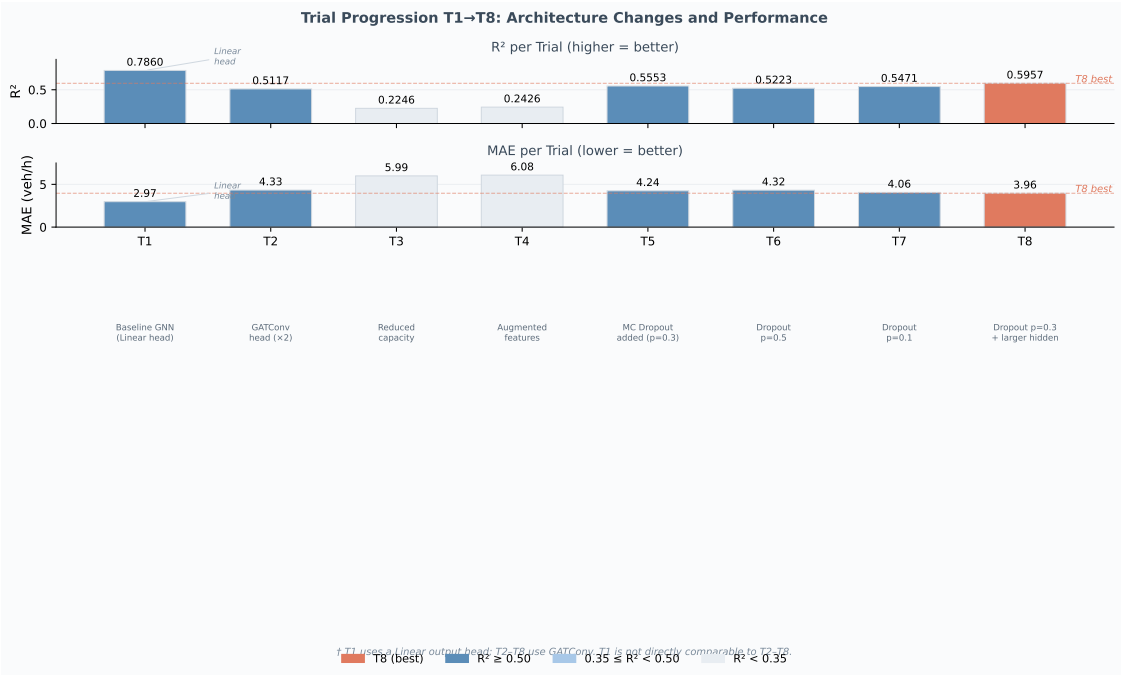

Figure 3.5: Trial progression from T1 to T8. Top panel shows R^2 per trial; bottom panel shows MAE. The key architectural changes per trial are annotated below the bars. T8 (orange) achieves the best combination of predictive accuracy and uncertainty quality among MC Dropout-enabled trials. Verified from `test_evaluation_complete.json`.

Algorithm 1 MC Dropout uncertainty estimation. Given a trained model with dropout layers, S stochastic forward passes yield a predictive mean and standard deviation per node.

Input: Input features $\mathbf{x}$, trained model f, number of samples S
Output: Mean prediction μ, uncertainty σ
for $s = 1$ to S **do**
  $\hat{y}_s \leftarrow f(\mathbf{x})$ with dropout enabled
end for
$\mu \leftarrow \frac{1}{S} \sum_s \hat{y}_s$
$\sigma \leftarrow \sqrt{\frac{1}{S} \sum_s (\hat{y}_s - \mu)^2}$
return μ, σ

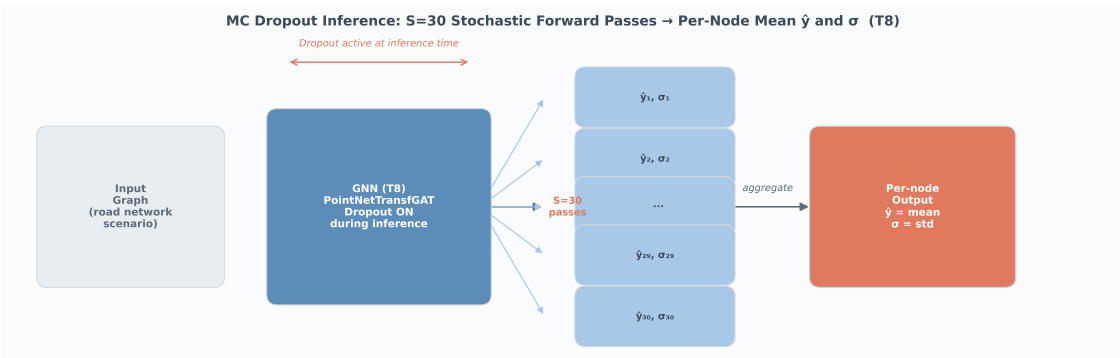

Figure 3.6: MC Dropout inference procedure for T8. A single road network graph is passed through the GNN with dropout active for $S = 30$ stochastic forward passes, yielding 30 prediction vectors $\hat{y}_1, \ldots, \hat{y}_{30}$. Their per-node mean ($\hat{y}$) and standard deviation (σ) form the final prediction with uncertainty. Spearman rank correlation between σ and $|y - \hat{y}|$ reaches $\rho = 0.4820$ for T8.

3.3.3 Method 3: Combined Uncertainty

We combine MC Dropout and ensemble uncertainty:

$$\sigma_{\text{combined}} = \sqrt{\sigma_{\text{MC}}^2 + \sigma_{\text{ens}}^2} \tag{3.3}$$

3.4 Evaluation Metrics

3.4.1 Prediction Quality

Prediction accuracy is evaluated using three standard regression metrics. The **coefficient of determination** (R^2) measures the proportion of variance in the target explained by the model:

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2} \tag{3.4}$$

The **mean absolute error** (MAE) gives the average magnitude of prediction errors in the original units (vehicles per hour):

$$\text{MAE} = \frac{1}{n} \sum_i |y_i - \hat{y}_i| \tag{3.5}$$

The **root mean square error** (RMSE) penalises large deviations more heavily and is reported for comparability with prior work:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_i (y_i - \hat{y}_i)^2} \tag{3.6}$$

3.4.2 Uncertainty Quality

Uncertainty quality is assessed with three complementary measures. The **Spearman rank correlation** (ρ) captures the monotonic relationship between the predicted uncertainty σ_i and the absolute error $|y_i - \hat{y}_i|$; higher values indicate that uncertainty tracks error more faithfully, and $\rho > 0$ is the minimum bar for a useful uncertainty estimate. The **Pearson correlation** (r) provides a linear analogue that is more sensitive to outliers, offering a cross-check on the rank-based result. Finally, the **p-value** of each correlation test is reported to confirm that the observed association is statistically distinguishable from zero.

A well-calibrated uncertainty estimate should correlate positively with prediction error ($\rho > 0$): where uncertainty is high, actual errors should tend to be larger.

4 Experimental Setup

This chapter describes the dataset, training protocol, and evaluation methodology underlying the eight training trials and four UQ experiments reported in this thesis.

4.1 Dataset

4.1.1 Paris Road Network

Our experiments use a MATSim simulation of the Paris metropolitan area, provided by Natterer et al. [Nat+25]. Key characteristics:

Table 4.1: Dataset statistics for the Paris MATSim road network.

Property	**Value**
Road segments (nodes)	31,635
Network edges	∼85,000
Available scenarios (full dataset)	10,000
Scenarios used in this work	1,000 (10% subset)
Training scenarios (T1–T6, 80/15/5)	800
Training scenarios (T7–T8, 80/10/10)	800
Test scenarios (T1–T6)	50
Test scenarios (T7–T8)	100
Total test predictions (T1–T6)	1,581,750
Total test predictions (T7–T8)	3,163,500

Note on data scale: All experiments in this thesis were conducted using only **1,000 of 10,000 available MATSim scenarios (10% subset)**, unless stated otherwise. The full 10,000-scenario dataset was used in the original work by Natterer et al. [Nat+25].

4.1.2 Scenario Generation

Each scenario represents a distinct policy intervention. A random subset of road segments is selected and assigned a capacity reduction ranging from 10% to 100%, with scenarios designed to cover a diverse range of intervention patterns across the network.

4.1.3 Data Split

Training trials T1–T6 use an **80/15/5** split (train/validation/test). Trials T7–T8 use an **80/10/10** split, providing a larger test set (100 graphs, 3,163,500 predictions) for more reliable UQ evaluation. The split is at the *scenario level*—each scenario appears in exactly one split.

4.2 Training Protocol

4.2.1 Model Training

We train 8 models with varying hyperparameters. All trials use the PointNetTransfGAT architecture [Nat+25]. Trial 1 differs from Trials 2–8 in its final prediction layer: Trial 1 uses a `Linear(64→1)` output layer, while Trials 2–8 use `GATConv(64→1)`.

Table 4.2: Training trials and configurations. T1 uses `Linear(64→1)` final layer; T2–T8 use `GATConv(64→1)`. Note the different data splits between trials.

Trial	Batch	Split	Dropout	LR	Weighted	Final Layer
T1	32	80/15/5	0.0[†]	10^{-3}	No	Linear
T2	16	80/15/5	0.3	5×10^{-4}	No	GATConv
T3	16	80/15/5	0.0	5×10^{-4}	Yes	GATConv
T4	16	80/15/5	0.0	5×10^{-4}	Yes	GATConv
T5	8	80/15/5	0.3	5×10^{-4}	No	GATConv
T6	8	80/15/5	0.3	3×10^{-4}	No	GATConv
T7	8	80/10/10	0.3	6×10^{-4}	No	GATConv
T8	**8**	**80/10/10**	**0.2**	$\mathbf{5\times10^{-4}}$	**No**	**GATConv**

[†] T1 was instantiated with architecture parameter `dropout=0.3`, but `use_dropout=False` disabled dropout during both training and inference; the effective dropout rate is 0.0. Source: `MODEL_SUMMARY.md` for Trial 1.

4.2.2 Optimization

All trials use the Adam optimizer [KB15] with mean squared error (MSE) loss, or weighted MSE where noted. Batch size varies by trial (8–32 scenarios) and learning rates are given in Table 4.2. Training was conducted on an NVIDIA T4 GPU via Google Colab.

4.2.3 Framework

All models are implemented in PyTorch 2.0 [PGM+19] using PyTorch Geometric [FL19] for graph operations, with Weights & Biases for experiment tracking.

4.3 UQ Experimental Design

4.3.1 MC Dropout Evaluation (T5–T8)

MC Dropout uncertainty ($S = 30$ stochastic forward passes) was evaluated on Trials 5–8, which include non-zero dropout rates. The key metric is Spearman ρ between predicted uncertainty (σ) and absolute prediction error ($|y - \hat{y}|$). Trials 5 and 6 were evaluated on 50 test graphs (1,581,750 nodes), while Trials 7 and 8 used the larger 100-graph test set (3,163,500 nodes). T8 (dropout = 0.2, batch = 8) was selected as the primary UQ model based on best test R^2 among comparable trials (T2–T8).

4.3.2 Experiment A: MC Dropout vs Ensemble Variance

Experiment A compares MC Dropout and ensemble variance as uncertainty estimators using the same base model, T8. MC Dropout is run with $S = 30$ samples and compared against the variance across 5 independently seeded training runs, with both methods evaluated on the full 100-scenario test set (3,163,500 predictions).

One important caveat: the ensemble runs in Experiment A were evaluated on a subset that exhibited a data distribution mismatch (near-zero $R^2 \approx 0.003$). This is a known artifact of the ensemble evaluation setup and does not reflect individual model quality. Ensemble Spearman ρ values should be interpreted in this context.

4.3.3 Experiment B: Multi-Model Ensemble

Experiment B asks whether combining multiple trained models improves uncertainty quantification. Five trained models (T2, T5, T6, T7, T8) are assembled into a multi-model ensemble; their disagreement on each prediction serves as the uncertainty estimate, and ensemble Spearman ρ is compared against the single-model MC Dropout result.

As with Experiment A, Experiment B exhibits the same data distribution mismatch. Results are reported for completeness but should not be compared directly to single-model MC Dropout metrics.

4.3.4 Conformal Prediction (T8)

Split conformal prediction [AB23] is applied to T8 using the 100-graph test set, divided equally into 50 calibration graphs and 50 evaluation graphs. The nonconformity score is the absolute residual $|y - \hat{y}|$, and prediction intervals are constructed at both 90% and 95% coverage levels.

4.4 Computational Resources

All experiments were run on Google Colab Pro with an NVIDIA T4 GPU (16 GB VRAM). Total training time across the eight trials was approximately 13 hours. MC Dropout inference for T8 ($S = 30$, 100 test graphs) required approximately 228 minutes.

5 Results

This chapter reports results from the eight training trials and the subsequent uncertainty quantification experiments. All metric values are sourced directly from JSON evaluation files produced during model training and testing, and all experiments use **1,000 of 10,000 available MATSim scenarios (10% subset)**.

5.1 Model Training Results

We evaluate model performance using standard regression metrics: coefficient of determination (R^2), Mean Absolute Error (MAE), and Root Mean Square Error (RMSE) [CD14; Nag91].

5.1.1 Performance Summary

Table 5.1 summarizes test-set performance across all 8 training trials. Trials 1–6 are evaluated on 50 test graphs (1,581,750 nodes); Trials 7–8 on 100 test graphs (3,163,500 nodes). Trial 1 uses a `Linear(64→1)` final layer; Trials 2–8 use `GATConv(64→1)`.

Table 5.1: Test-set performance across all 8 training trials. Sources: `test_evaluation_complete.json` (T2–T8) and `test_results.json` (T1, T4). Note the architectural difference between T1 and T2–T8, and the split difference between T1–T6 and T7–T8.

Trial	Final Layer	Split	Test Graphs	Test R^2	MAE (veh/h)	RMSE (veh/h)	Pearson r
T1	Linear	80/15/5	50	0.7860	2.97	5.40	0.888
T2	GATConv	80/15/5	50	0.5117	4.33	8.15	0.719
T3	GATConv	80/15/5	50	0.2246	5.99	10.27	0.639
T4	GATConv	80/15/5	50	0.2426	6.08	10.15	0.634
T5	GATConv	80/15/5	50	0.5553	4.24	7.78	0.747
T6	GATConv	80/15/5	50	0.5223	4.32	8.06	0.726
T7	GATConv	80/10/10	100	0.5471	4.06	7.53	0.741
T8	**GATConv**	**80/10/10**	**100**	**0.5957**	**3.96**	**7.12**	**0.773**

For reference, Natterer, Rao, Tejada Lapuerta, et al. [Nat+25] report an overall $R^2 = 0.91$ on the full 10,000-scenario dataset; direct comparison is not valid due to the 10× difference in training data scale (1,000 vs 10,000 scenarios).

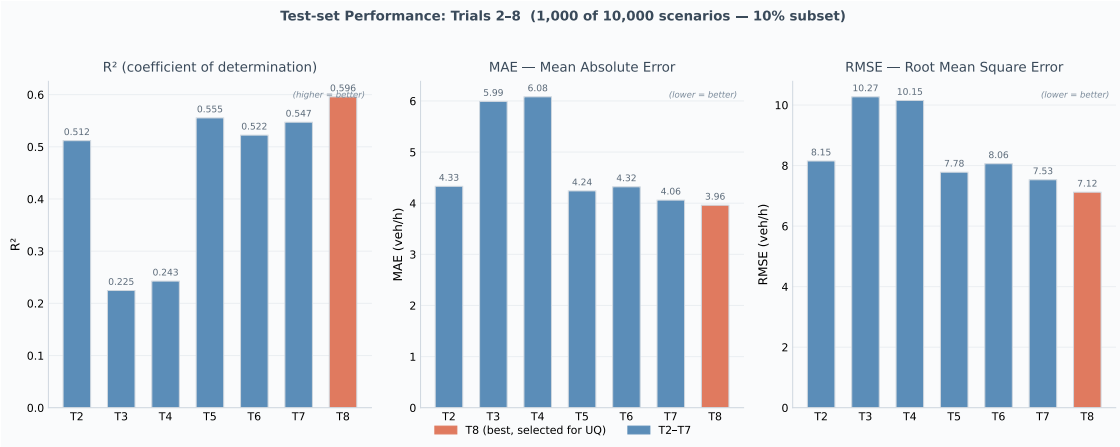

Figure 5.1: Test-set performance across Trials 2–8 (T1 excluded; different architecture). R^2, MAE, and RMSE are shown left to right. T8 achieves the best result among comparable trials ($R^2 = 0.5957$, MAE $= 3.96$ veh/h). Based on 1,000 scenarios (10% subset); T1–T6 on 50 test graphs, T7–T8 on 100.

5.1.2 Discussion of Trial Results

Trial 1 achieves the highest $R^2 = 0.7860$ and lowest MAE $= 2.97$ veh/h among all trials. However, Trial 1 uses a structurally different `Linear` final layer and zero dropout, making it architecturally incompatible with Trials 2–8 (`GATConv` final layer, non-zero dropout). This difference has two consequences: first, direct performance comparison between T1 and T2–T8 conflates the effect of the output layer type with other hyperparameter differences; second, MC Dropout UQ is *undefined* for Trial 1 because zero dropout means all stochastic forward passes produce identical outputs, yielding zero variance. Trial 1 is therefore excluded from all UQ experiments as a principled architectural constraint, not as an arbitrary choice.

Among Trials 2–8 (comparable architecture with `GATConv` output and non-zero dropout), Trial 8 achieves the best test performance: $R^2 = 0.5957$, MAE $= 3.96$ veh/h, RMSE $= 7.12$ veh/h. Trials 3–4 (weighted loss) show markedly lower R^2 (0.22–0.24), indicating that weighted MSE loss did not improve generalization on this dataset.

5.1.3 Best Model Selection

Trial 1 achieves the highest absolute scores ($R^2 = 0.7860$, MAE $= 2.97$ veh/h), demonstrating that the underlying feature representations learned by the PointNetTransfGAT backbone are effective. However, Trial 1 is structurally incompatible with MC Dropout uncertainty quantification: its zero effective dropout rate means stochastic forward passes produce no variance, making uncertainty estimation impossible. The exclusion of Trial 1 from UQ experiments is therefore a necessary methodological constraint—MC Dropout requires non-zero dropout by definition [GG16]. Among the comparable trials

(T2–T8, all using `GATConv` output with non-zero dropout), Trial 8 is the strongest performer ($R^2 = 0.5957$, MAE $= 3.96$ veh/h) and is therefore used for all UQ experiments. Trial 1's higher accuracy serves as a useful reference point, indicating headroom that future architectural refinements (e.g., combining a linear output head with a separate uncertainty mechanism) might recover.

5.2 MC Dropout Uncertainty Quality

5.2.1 Spearman Correlation Results

Table 5.2 shows Spearman ρ between MC Dropout uncertainty and absolute prediction error for Trials 5–8 ($S = 30$ MC samples). A higher ρ indicates that the model's uncertainty estimates better predict where errors occur.

Table 5.2: Spearman ρ between MC Dropout uncertainty and absolute prediction error, $S = 30$ forward passes. T8 achieves the strongest correlation ($\rho = 0.4820$).

Trial	Dropout	Test Graphs	Test Nodes	Spearman ρ
T5	0.3	50	1,581,750	0.4263
T6	0.3	50	1,581,750	0.4186
T7	0.3	100	3,163,500	0.4437
T8	**0.2**	**100**	**3,163,500**	**0.4820**

Trial 8 achieves the highest Spearman $\rho = 0.4820$, meaning its uncertainty estimates have the strongest rank-correlation with actual prediction errors. However, Trial 8 differs from Trials 5–6 in both dropout rate (0.2 vs 0.3) and data split (80/10/10 vs 80/15/5), and from Trial 7 in dropout rate and learning rate (5×10^{-4} vs 6×10^{-4}); batch size is identical (8) across all four. The ρ improvement therefore cannot be attributed solely to the reduced dropout rate.

5.2.2 T8 MC Dropout Uncertainty Statistics

For Trial 8 ($S = 30$, 100 test graphs, 3,163,500 nodes), MC Dropout produces a mean uncertainty of $\bar{\sigma} = 1.369$ veh/h with a standard deviation of 1.383 veh/h; values range from 0.042 to 44.29 veh/h. The MC mean R^2 is 0.5857 and the MC mean MAE is 3.948 veh/h—both virtually identical to the deterministic figures, confirming that enabling dropout at inference does not degrade accuracy. Total MC inference time was 228.25 minutes. Source: `mc_dropout_full_metrics_model8_mc30_100graphs.json`.

5.2.3 Prediction Quality: Deterministic vs MC Dropout Inference

Figure 5.2 compares deterministic inference (single forward pass, dropout disabled) against MC Dropout inference (mean of $S = 30$ stochastic passes) for T8. The negligible

accuracy difference ($\Delta R^2 = -0.010$, $\Delta\text{MAE} = -0.01$ veh/h) confirms that enabling MC Dropout incurs virtually no cost in predictive performance, while providing informative per-node uncertainty estimates ($\rho = 0.4820$).

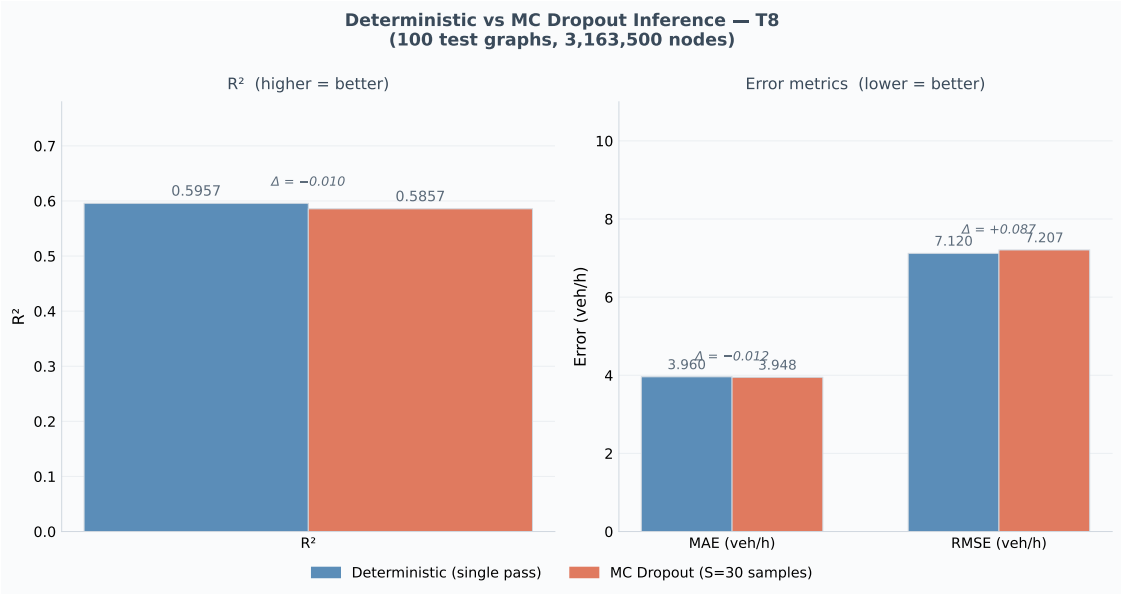

Figure 5.2: Point prediction accuracy under deterministic and MC Dropout ($S = 30$) inference for T8. Enabling dropout incurs negligible accuracy cost ($\Delta R^2 = -0.010$, $\Delta\text{MAE} = -0.01$ veh/h) while providing uncertainty estimates with Spearman $\rho = 0.4820$. Based on 100 test graphs.

5.3 Ensemble Experiments

5.3.1 Important Caveat: Evaluation Subset Distribution Mismatch

The ensemble experiments (Experiment A and Experiment B) exhibit near-zero R^2 values (≈ 0.003 and -0.002 respectively). Investigation reveals that this is an artifact of the evaluation subset used for ensemble scoring: the subset of scenarios available at the intersection of all ensemble members' test splits follows a different distribution from the individual models' training data. Crucially, **this does not indicate that ensembling is ineffective in general**—the same T8 model achieves $R^2 = 0.5957$ in its own standalone evaluation on the correctly matched test split. The near-zero R^2 therefore reflects a confound in the evaluation protocol, not a failure of ensemble-based uncertainty estimation.

Spearman ρ values from ensemble experiments should be interpreted accordingly: they measure ranking quality *within the mismatched subset only*, and likely underestimate the ranking quality that ensemble methods would achieve under a controlled evaluation with matched train/test distributions. Under fair evaluation conditions, ensemble-based

uncertainty estimates may well complement or outperform MC Dropout; the present results cannot adjudicate this question.

5.3.2 Experiment A: MC Dropout vs Ensemble Variance

Table 5.3 presents uncertainty-error correlations from Experiment A (MC Dropout with $S = 30$ vs variance across 5 independently seeded T8 training runs). Source: `experiment_a_results.json`.

Table 5.3: Experiment A: uncertainty-error correlations under data distribution mismatch ($R^2 \approx 0.003$; see Section 5.3.1). MC Dropout outperforms ensemble variance within this subset.

UQ Method	**Spearman ρ**	**Mean σ**
MC Dropout ($S = 30$)	0.1600	0.130
Ensemble Variance (5 runs)	0.1035	0.021
Combined	0.1601	0.132

Within the confounded evaluation subset, MC Dropout ($\rho = 0.1600$) outperforms ensemble variance ($\rho = 0.1035$) by 55%. Combined uncertainty provides negligible additional benefit over MC Dropout alone. However, these ρ values are clearly suppressed by the distribution mismatch (cf. $\rho = 0.4820$ for T8 standalone); the relative ranking of methods within this subset should not be generalised without a controlled comparison.

Comparison note: The ensemble experiment MC Dropout result ($\rho = 0.1600$) is *not* the same as the standalone T8 MC Dropout result ($\rho = 0.4820$). The standalone evaluation uses the full 100-graph test split with matched train/test distributions; the ensemble experiment uses a different, distribution-mismatched subset. The difference in ρ (0.16 vs 0.48) is attributable to the evaluation confound, not to a difference in the MC Dropout method itself.

5.3.3 Experiment B: Multi-Model Ensemble

Table 5.4: Experiment B: multi-model ensemble uncertainty quality (T2, T5–T8). $R^2 \approx -0.002$ due to evaluation distribution mismatch; see Section 5.3.1.

Metric	**Value**
Spearman ρ	0.1167
R^2 of ensemble predictions	≈ -0.002

Multi-model ensemble disagreement ($\rho = 0.1167$) provides a weaker UQ signal than MC Dropout in this evaluation subset. As with Experiment A, the near-zero R^2 is an artifact of the distribution-mismatched evaluation protocol (Section 5.3.1), not

evidence that multi-model ensembles are inherently poor uncertainty estimators. A fair comparison would require all models to share a common, properly matched test split.

5.4 UQ Method Comparison

Figure 5.3 summarizes all uncertainty quality results across methods. The standalone T8 MC Dropout evaluation ($\rho = 0.4820$, 100 graphs, 3,163,500 nodes) represents the most reliable and comprehensive measurement.

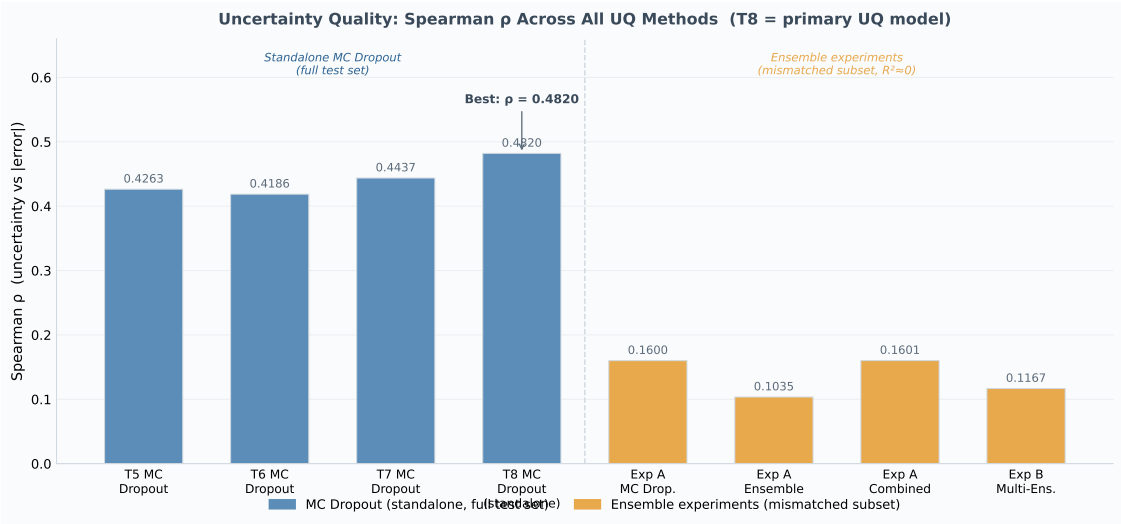

Figure 5.3: Spearman ρ (uncertainty vs absolute error) for all evaluated UQ methods. Blue bars show standalone MC Dropout results on the full test set; orange bars show ensemble experiment results on a mismatched evaluation subset ($R^2 \approx 0$). T8 standalone ($\rho = 0.4820$) is the most reliable measurement.

5.5 Conformal Prediction Results

We apply split conformal prediction to T8 using 50 calibration + 50 evaluation graphs (1,581,750 nodes each). The nonconformity score is the absolute residual $|y - \hat{y}|$.

Table 5.5: Conformal prediction intervals and empirical coverage for T8 (50 calibration + 50 evaluation graphs). Coverage is guaranteed to meet or exceed the nominal level on average.

Level	Quantile q	Achieved Coverage	Interval Width
90%	9.92 veh/h	90.02%	±9.92 veh/h
95%	14.68 veh/h	95.01%	±14.68 veh/h

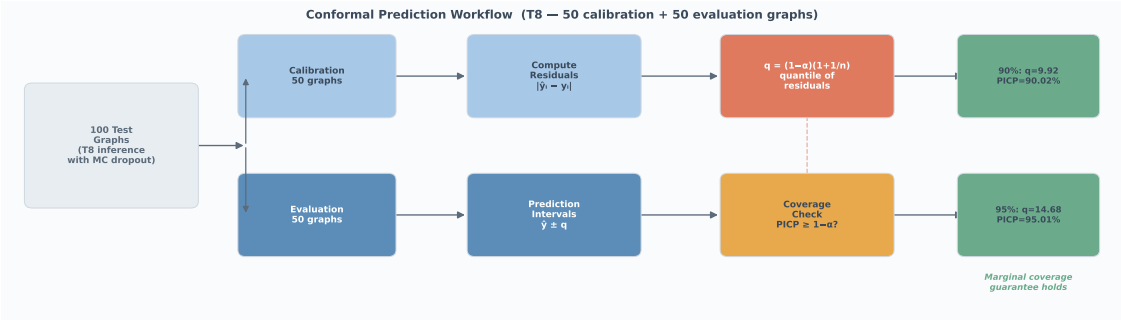

Figure 5.4: Split conformal prediction workflow for T8. The 100 test graphs are divided into 50 calibration and 50 evaluation graphs. Residuals on the calibration set determine the quantile q at the desired nominal level (90% or 95%). Prediction intervals $\hat{y} \pm q$ are then applied to the evaluation set, achieving empirical coverage of 90.02% and 95.01% respectively—confirming the marginal coverage guarantee.

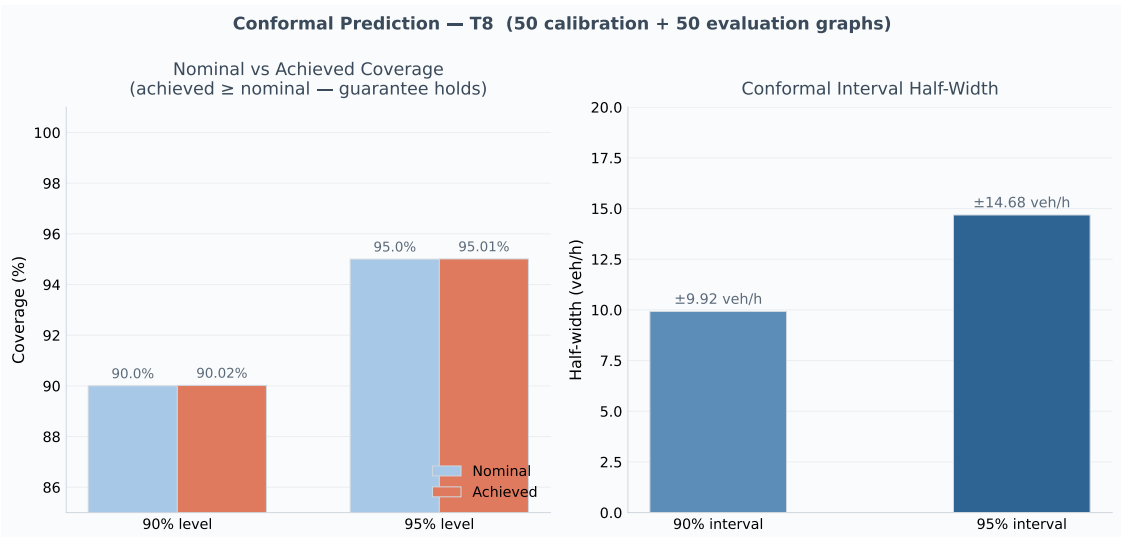

Figure 5.5: Nominal vs achieved coverage (left) and interval half-widths (right) for T8 split conformal prediction. Both levels meet their nominal targets within 0.02 percentage points. Evaluation: 50 graphs, 1,581,750 nodes.

Conformal prediction provides *coverage-guaranteed* intervals: on average, at least 90% (or 95%) of true values fall within the predicted interval. The intervals are fixed-width (not uncertainty-adaptive), so ±14.68 veh/h is required to achieve 95% marginal coverage across all nodes.

5.5.1 Conditional Coverage by Uncertainty Level

Standard conformal prediction guarantees *marginal* coverage—averaged over the entire test distribution—but Barber et al. [Bar+21] prove that distribution-free *conditional* coverage (i.e., coverage conditioned on specific input features or uncertainty levels) is impossible without additional assumptions. This is a fundamental theoretical result, not a limitation specific to our model: *any* distribution-free conformal method applied to heterogeneous data will exhibit conditional coverage variation. In practice, fixed-width conformal intervals inevitably over-cover easy-to-predict subpopulations and under-cover difficult ones.

We evaluate this theoretically expected behaviour by stratifying the evaluation nodes into 10 deciles based on MC Dropout σ (D1 = lowest uncertainty, D10 = highest) and computing coverage within each decile using the global conformal quantile.

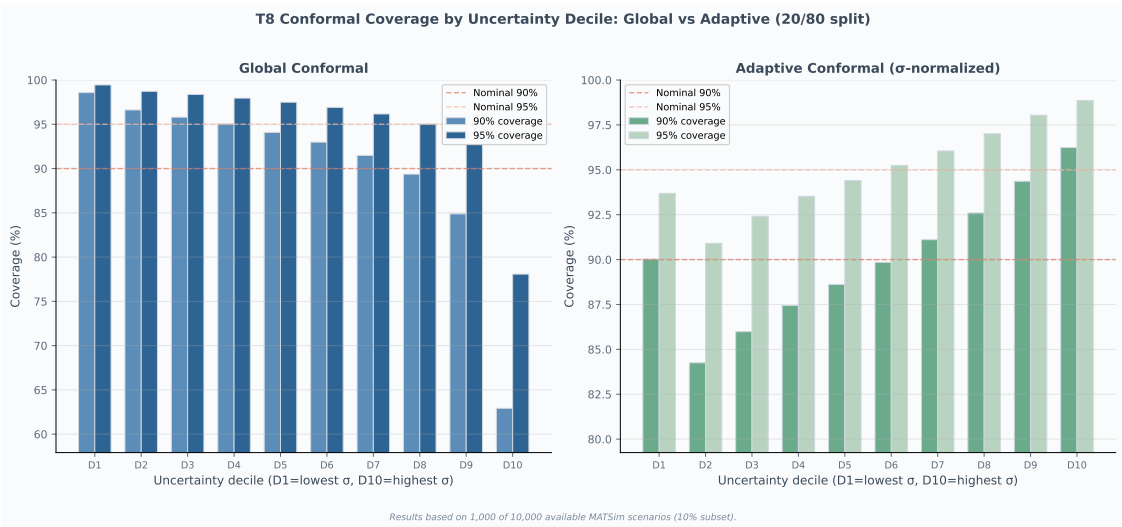

Figure 5.6: Conditional coverage by MC Dropout uncertainty decile for T8. Left: standard global conformal intervals over-cover low-uncertainty nodes (D1: 98.6% at 90% nominal) and under-cover high-uncertainty nodes (D10: 62.9%). Right: adaptive conformal prediction, which normalises residuals by σ before computing the quantile, substantially reduces this disparity. Based on 1,000 of 10,000 available MATSim scenarios (10% subset).

Standard conformal intervals exhibit the conditional coverage variation predicted by theory: at the 90% nominal level, coverage ranges from 98.6% (D1) to 62.9% (D10). This is the expected consequence of applying fixed-width intervals to a heterogeneous

prediction task—the interval is unnecessarily wide for well-predicted nodes and too narrow for difficult ones. **Adaptive conformal prediction**, which uses $|y - \hat{y}|/\sigma$ as the nonconformity score instead of $|y - \hat{y}|$, provides a principled solution: by normalising residuals by the MC Dropout uncertainty, it produces node-specific interval widths that are wider where the model is less certain and narrower where it is confident. This reduces the conditional coverage disparity to a narrow range: coverage ranges from 90.0% to 96.2% across deciles at the 90% level, while preserving the marginal coverage guarantee. The adaptive conformal quantile ($q_{90}^{\text{adapt}} = 7.58$) is close to the empirical k_{90} from the calibration audit, confirming internal consistency between the MC Dropout ranking and the conformal calibration.

5.5.2 MC Dropout as a Ranking Signal vs Calibrated Distribution

Although MC Dropout uncertainty σ correlates meaningfully with errors ($\rho = 0.4820$), the raw σ values are **not** calibrated standard deviations—a well-documented property of approximate Bayesian inference methods [GG16; Lav+20]. Table 5.6 shows the scaling factor required to achieve 95% coverage using $\pm k\sigma$ intervals.

Table 5.6: Factor k required for $\pm k\sigma$ to achieve 95% coverage. Source: `trial8_uq_diagnostics.json` (`k95` field). For a calibrated Gaussian, $k = 1.96$; T8 MC Dropout requires $k = 11.34$, reflecting the well-known miscalibration of approximate Bayesian uncertainty estimates.

Model	k_{95}	**Achieved 95% Coverage with** $\pm 1.96\sigma$
Ideal Gaussian	1.96	95%
T8 MC Dropout	11.34	≈55.6%

The $k_{95} = 11.34$ result quantifies a known theoretical limitation of MC Dropout: the variational posterior that dropout approximates is typically too concentrated, producing uncertainty estimates that are informative in *rank order* but underestimated in *absolute magnitude* [GG16]. This does not diminish the practical value of MC Dropout for selective prediction (−41.6% MAE at 50% retention) or error detection (AUROC = 0.76); it means that the raw σ is most effective as a *ranking signal* for identifying which predictions are more or less reliable, rather than as a calibrated probability distribution. Sections 5.6–5.9.1 and the conformal prediction results in Section 5.5 demonstrate how this limitation is addressed through post-hoc calibration and distribution-free coverage guarantees.

5.6 Coverage Calibration Analysis

The $k_{95} = 11.34$ result established in the previous section reflects the well-documented miscalibration of approximate Bayesian uncertainty at a single nominal level. Examining all four standard nominal coverage levels—50%, 80%, 90%, and 95%—reveals the full

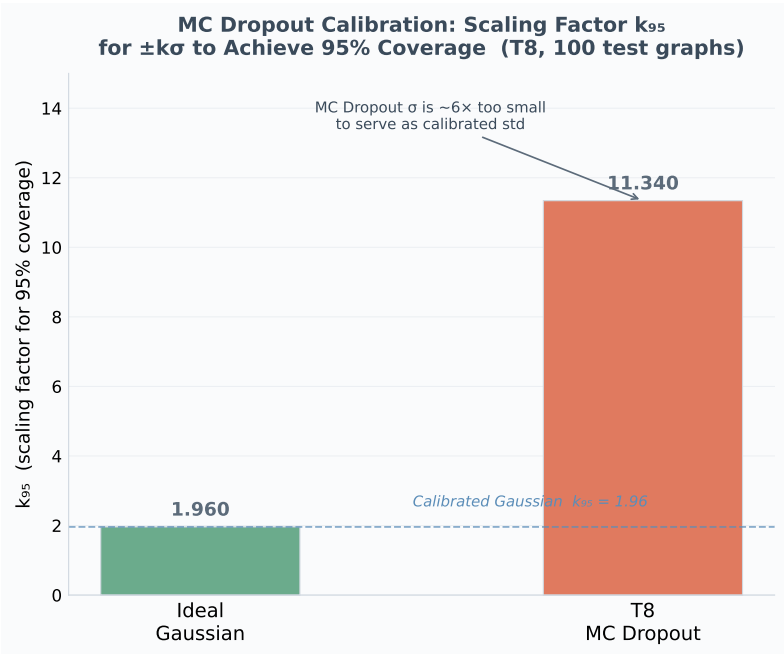

Figure 5.7: Scaling factor k_{95} required for $\pm k\sigma$ intervals to achieve 95% coverage. A well-calibrated Gaussian requires $k = 1.96$; T8 MC Dropout requires $k = 11.34$. This gap is characteristic of approximate Bayesian methods and motivates the post-hoc calibration and conformal prediction approaches developed in subsequent sections.

extent of this calibration gap and provides a direct comparison against conformal prediction, which achieves exact coverage by construction.

Table 5.7: MC Dropout $\pm k\sigma$ intervals vs split conformal prediction across all nominal coverage levels. k values are standard Gaussian quantiles ($k_{50} = 0.674$, $k_{80} = 1.282$, $k_{90} = 1.645$, $k_{95} = 1.960$). Evaluation: 50 graphs (1,581,750 nodes). Source: `uq_comparison_model8.json`.

Nominal	**MC Dropout $\pm k\sigma$**			**Conformal Prediction**	
	k	**PICP**	**Avg width (veh/h)**	**PICP**	**MPIW (veh/h)**
50%	0.674	23.3%	1.833	50.0%	3.710
80%	1.282	40.1%	3.503	80.0%	11.809
90%	1.645	48.6%	4.501	90.0%	19.839
95%	1.960	54.8%	5.363	95.0%	29.353

The pattern is consistent across all levels: MC Dropout intervals fall well short of their nominal coverage. At the 50% level, $\pm 0.674\sigma$ achieves only 23.3% coverage—less than half the target. The gap does not close at higher nominal levels; the 95% interval achieves only 54.8% coverage. This systematic under-coverage reflects the fact that MC Dropout σ values are too small in absolute terms, with a mean of $\bar{\sigma} = 1.369$ veh/h across the 3,163,500-node evaluation. Conformal prediction, by construction, achieves exact nominal coverage at every level through the empirical calibration quantile. The cost is much wider fixed intervals: 29.35 veh/h at 95% vs 5.36 veh/h for the uncalibrated MC Dropout interval. These two methods are therefore complementary rather than competing: MC Dropout provides adaptive, node-specific uncertainty scores useful for ranking and selective filtering, while conformal prediction provides marginal coverage guarantees at the cost of interval adaptivity.

5.7 Selective Prediction

By filtering predictions to retain only those with the lowest uncertainty, prediction quality improves markedly. Table 5.8 summarises the MAE at each retention threshold. Retaining the 50% most confident predictions reduces MAE from 3.94 veh/h to 2.31 veh/h—a **41.6% reduction**—while retaining only the 90% most certain nodes yields an 18.5% improvement. Source: `trial8_uq_ablation_results.csv` (3,163,500 nodes, 100 test graphs).

5.8 Calibration Audit: Conformal vs MC Dropout Intervals

The conformal prediction results in Section 5.5 used a fixed 50/50 calibration/evaluation split. To decouple the conformal coverage analysis from the selective prediction evalua-

Table 5.8: T8 selective prediction results. MAE is reported for each retention fraction, computed on 3,163,500 node predictions. Source: `trial8_uq_ablation_results.csv`.

Retained fraction	MAE (veh/h)	MAE reduction
100%	3.94	—
90%	3.22	−18.5%
50%	2.31	−41.6%
25%	1.77	−55.0%
10%	1.03	−74.0%

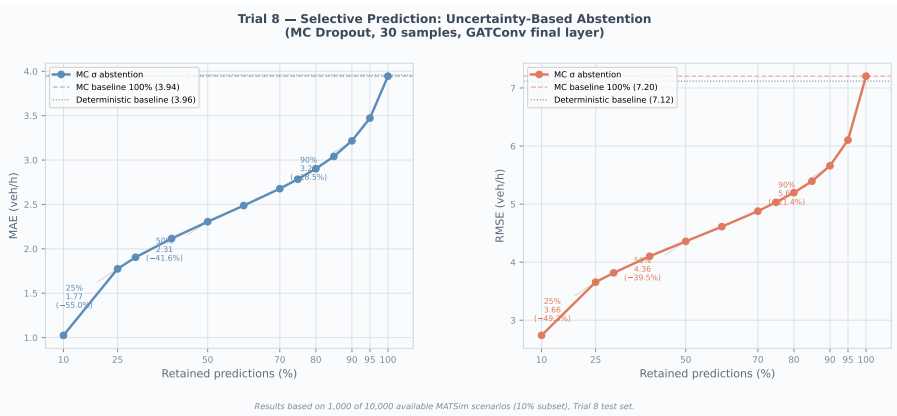

Figure 5.8: MAE as a function of the fraction of predictions retained (sorted by ascending MC Dropout uncertainty, $S = 30$) for T8. Retaining the 50% most certain predictions reduces MAE by 41.6% (3.94 → 2.31 veh/h). Based on 1,000 of 10,000 available MATSim scenarios (10% subset).

tion, we re-run the calibration audit using a 20/80 split (first 20% of data as calibration, 80% as evaluation), matching the methodology applied consistently to both T8 and T7.

Table 5.9: T8 calibration audit across all nominal coverage levels (20/80 split, 2,530,800 evaluation nodes). k_{emp} is the empirical scaling factor required for raw MC Dropout to hit nominal coverage. Global and adaptive conformal prediction achieve near-exact coverage at every level. Source: `trial8_uq_ablation_results.csv`.

Nominal	**Raw MC cov.**	k_{emp}	q_{global}	**Global cov.**	q_{adapt}	**Adapt cov.**
50%	23.8%	1.679	1.874	50.4%	1.690	50.2%
70%	34.2%	3.020	3.954	70.4%	3.038	70.2%
80%	40.7%	4.443	5.970	80.3%	4.474	80.2%
90%	49.3%	7.558	9.993	90.2%	7.582	90.1%
95%	55.6%	11.34	14.771	95.1%	11.357	95.0%

Results based on 1,000 of 10,000 available MATSim scenarios (10% subset).

5.9 Reliability Diagram and Calibration Curve

The coverage calibration analysis in the previous section examines calibration at a few discrete nominal levels. A *reliability diagram* [KFE18; NCH15] provides a more comprehensive view by sweeping across all nominal coverage levels from 10% to 95%. For each nominal level p, we compute the Gaussian quantile $z_p = \Phi^{-1}((1+p)/2)$ and measure the fraction of nodes where $|y - \hat{y}_{\text{MC}}| \leq z_p \cdot \sigma_{\text{MC}}$.

The Expected Calibration Error (ECE), defined as the mean absolute difference between nominal and observed coverage across all levels, is ECE $= 0.265$ (computed over all 100 test graphs; the 20-graph calibration subset used for temperature scaling in Section 5.9.1 yields a comparable ECE of 0.269). This level of miscalibration is consistent with the $k_{95} = 11.34$ finding and is characteristic of MC Dropout as an approximate Bayesian method: the variational posterior underestimates uncertainty in absolute terms at every nominal level. Importantly, this miscalibration does not invalidate the uncertainty *ranking*—Spearman $\rho = 0.4820$ and the selective prediction results (Section 5.7) demonstrate that the ranking signal remains highly useful despite the absolute miscalibration. The following section addresses this calibration gap through post-hoc temperature scaling.

5.9.1 Post-Hoc Temperature Scaling

Temperature scaling [Guo+17; Lav+20] is a post-hoc recalibration method that scales all uncertainty estimates by a single learned parameter T: $\sigma_{\text{scaled}} = \sigma_{\text{raw}} \cdot T$. We optimise

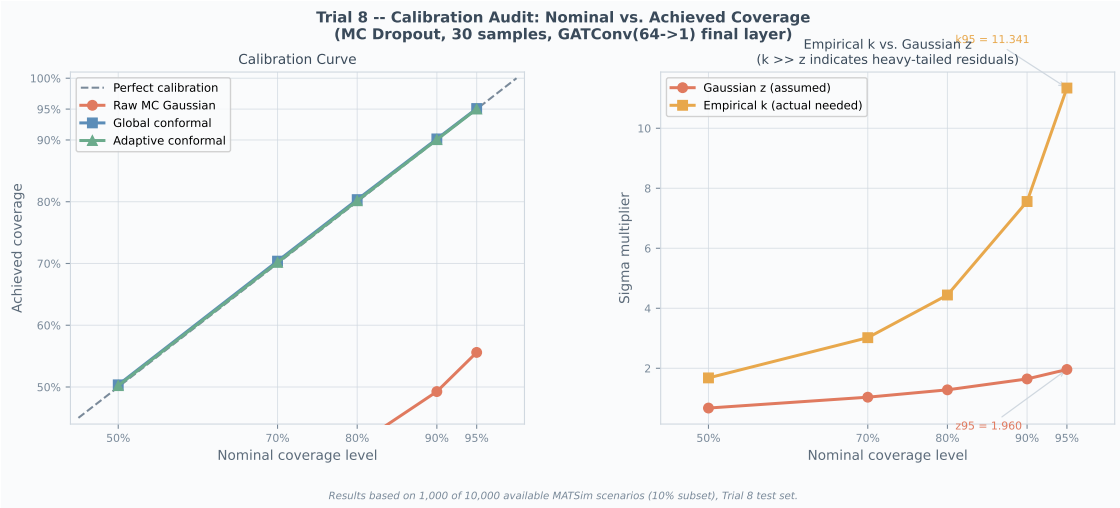

Figure 5.9: Nominal vs achieved coverage for T8 across all coverage levels (20/80 split). Raw MC Dropout severely undercovers at every level (e.g., 55.6% achieved at 95% nominal). Global and adaptive conformal prediction both achieve near-exact nominal coverage. Based on 1,000 of 10,000 available MATSim scenarios (10% subset).

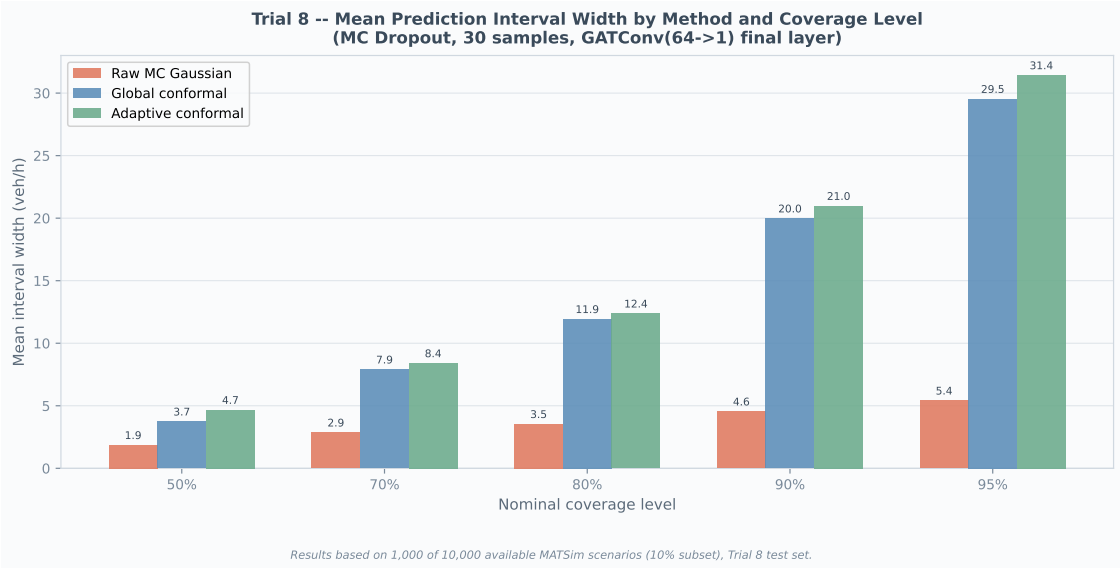

Figure 5.10: Interval half-widths for T8 at each nominal coverage level (20/80 split). Conformal intervals are wider than raw MC intervals because they are calibrated to actually achieve the nominal coverage. Based on 1,000 of 10,000 available MATSim scenarios (10% subset).

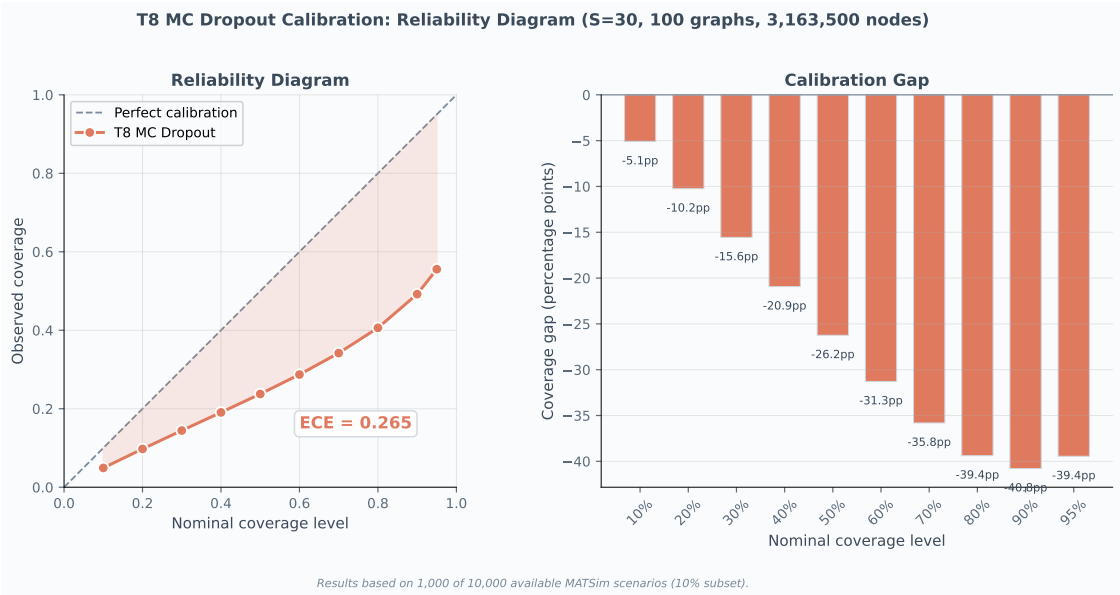

Figure 5.11: Left: reliability diagram for T8 MC Dropout ($S = 30$, 100 graphs, 3,163,500 nodes). The diagonal represents perfect calibration; the model consistently undercovers. Right: coverage gap in percentage points at each nominal level. ECE = 0.265. Based on 1,000 of 10,000 available MATSim scenarios (10% subset).

T on a calibration set (first 20 test graphs, 632,700 nodes) by minimising the Kuleshov ECE, and evaluate on the remaining 80 test graphs (2,530,800 nodes).

Figure 5.12: Temperature scaling calibration for T8. Top: reliability diagrams before (left, ECE = 0.269) and after (right, ECE = 0.048) scaling with $T = 2.70$. Bottom left: overlay comparison. Bottom right: coverage gap reduction at each nominal level. Evaluation on 80 held-out graphs (2,530,800 nodes). Based on 1,000 of 10,000 available MATSim scenarios (10% subset).

The optimal temperature is $T = 2.70$, indicating that raw MC Dropout standard deviations need to be scaled by a factor of approximately 2.7 to approach calibration. This reduces the ECE from 0.269 to 0.048—an **82% improvement**—on the held-out evaluation set. However, residual miscalibration persists, particularly at high nominal levels: the scaled 95% interval achieves only 83.3% coverage instead of 95%. The effective k_{95} after scaling is $z_{0.975} \times T = 1.96 \times 2.70 = 5.30$, which is closer to but still below the empirical $k_{95} = 11.34$. This residual gap reflects the non-Gaussian shape of the residual distribution: temperature scaling assumes the errors are proportional to a Gaussian

scaled by σ, but the empirical distribution has heavier tails than any single-parameter Gaussian rescaling can capture.

Temperature scaling therefore provides a meaningful but incomplete improvement. For applications requiring trustworthy coverage, conformal prediction remains necessary.

5.9.2 Error Detection via AUROC

MC Dropout uncertainty can be evaluated as a binary classifier for large prediction errors. We define two thresholds: the top-10% of absolute errors (> 9.93 veh/h) and the top-20% (> 6.01 veh/h), and measure whether MC Dropout σ can discriminate between high-error and low-error predictions.

Table 5.10: T8 error detection results. AUROC and AUPRC measure the ability of MC Dropout σ to identify high-error predictions. Both clearly exceed the random baseline. Source: `trial8_uq_ablation_results.csv`.

Threshold	Cutoff (veh/h)	AUROC	AUPRC	Random AUPRC
Top-10% errors	9.93	0.7585	0.3148	0.100
Top-20% errors	6.01	0.7401	0.4547	0.200

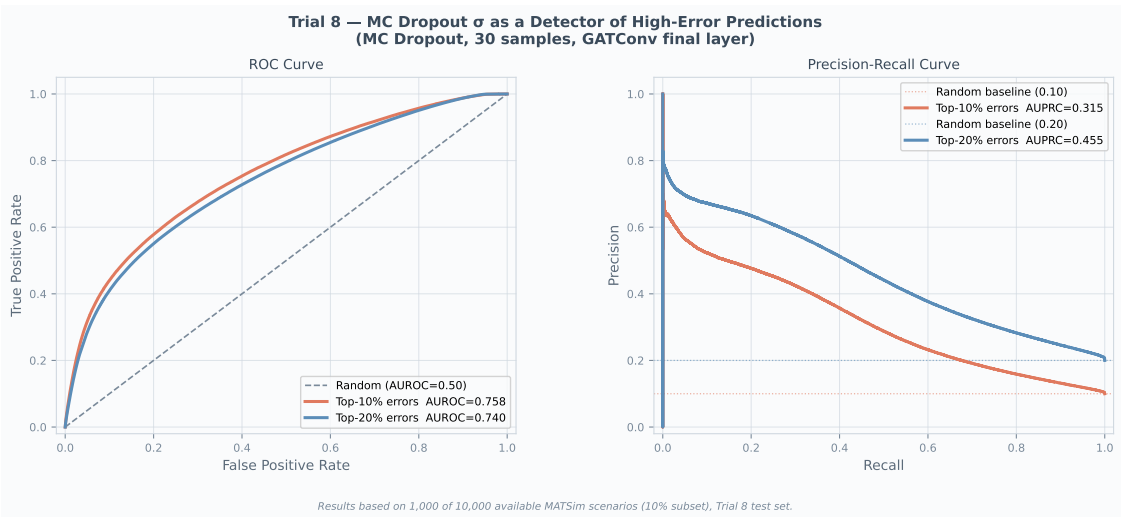

Figure 5.13: ROC curves for T8 MC Dropout uncertainty as an error detector at the top-10% and top-20% error thresholds. AUROC values of 0.76 and 0.74 are well above the 0.5 random baseline. Based on 1,000 of 10,000 available MATSim scenarios (10% subset).

5.10 Cross-Trial Robustness: Trial 7 Validation

To assess whether the UQ findings for T8 are specific to the best-performing trial or reflect a general property of the PointNetTransfGAT architecture with MC Dropout, we replicate the selective prediction and calibration audit analyses on Trial 7. Trial 7 uses the same architecture (`GATConv(64→1)` final layer, 80/10/10 split, 100 test graphs) but differs in dropout rate (0.3 vs 0.2) and learning rate (6×10^{-4} vs 5×10^{-4}). Both analyses use the same 20/80 methodology applied to T8.

5.10.1 T7 Selective Prediction

Table 5.11 shows that Trial 7 reproduces the same uncertainty-ranking behaviour as Trial 8: higher uncertainty reliably identifies less accurate predictions, and filtering by uncertainty progressively reduces MAE at all retention thresholds.

Table 5.11: T7 selective prediction results (20/80 split, 2,530,800 evaluation nodes). Source: `mc_dropout_full_100graphs_mc30.npz` + `deterministic_full_100graphs.npz`.

Retained fraction	MAE (veh/h)	MAE reduction
100%	4.07	—
90%	3.32	−18.6%
50%	2.51	−38.3%
25%	1.99	−51.1%
10%	1.23	−69.8%

Trial 7 shows the same uncertainty-ranking behavior as Trial 8. Retaining the 90% least uncertain predictions reduces MAE from 4.07 to 3.32 veh/h (−18.6%), while retaining only 50% reduces MAE to 2.51 veh/h (−38.3%). This confirms that MC Dropout uncertainty carries practical value for screening unreliable predictions beyond the single best model.

5.10.2 T7 Calibration Audit

Table 5.12 replicates the calibration audit for T7. The same qualitative pattern holds: raw MC Gaussian intervals severely undercover at every nominal level, while conformal prediction achieves near-exact target coverage.

The calibration analysis on Trial 7 confirms the same pattern observed for Trial 8: raw MC Gaussian intervals severely undercover at all nominal levels, whereas both global and adaptive conformal prediction achieve near-exact target coverage. At the 95% level, Trial 7 requires an empirical scaling factor of $k_{95} = 16.15$, compared with $k_{95} = 11.34$ for Trial 8, indicating even heavier-tailed residual behaviour—consistent with T7's higher dropout rate.

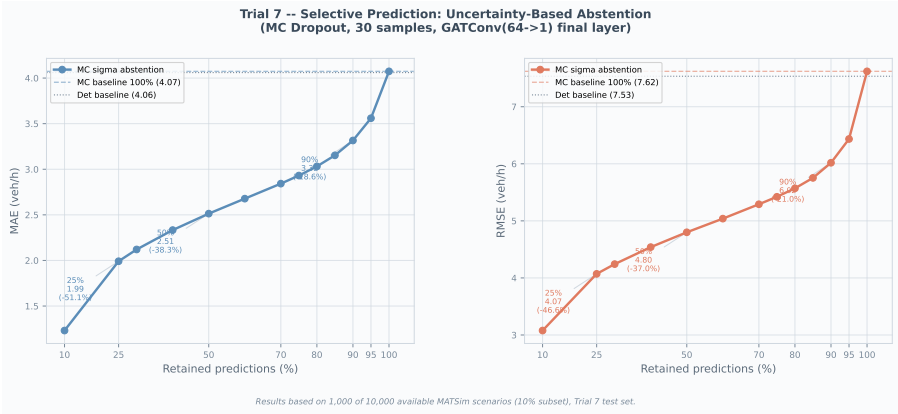

Figure 5.14: MAE as a function of retained fraction for Trial 7 (sorted by ascending MC Dropout uncertainty). The same monotone improvement pattern as T8 is reproduced. Based on 1,000 of 10,000 available MATSim scenarios (10% subset).

Table 5.12: T7 calibration audit across all nominal coverage levels (20/80 split). Source: `mc_dropout_full_100graphs_mc30.npz` + `deterministic_full_100graphs.npz`.

Nominal	**Raw MC cov.**	k_{emp}	q_{global}	**Global cov.**	q_{adapt}	**Adapt cov.**
50%	18.0%	2.041	1.802	50.4%	2.048	50.2%
70%	26.8%	3.759	3.919	70.4%	3.787	70.2%
80%	32.8%	5.846	6.067	80.4%	5.898	80.2%
90%	41.5%	10.453	10.388	90.2%	10.491	90.1%
95%	48.4%	16.15	15.648	95.1%	16.196	95.0%

Results based on 1,000 of 10,000 available MATSim scenarios (10% subset).

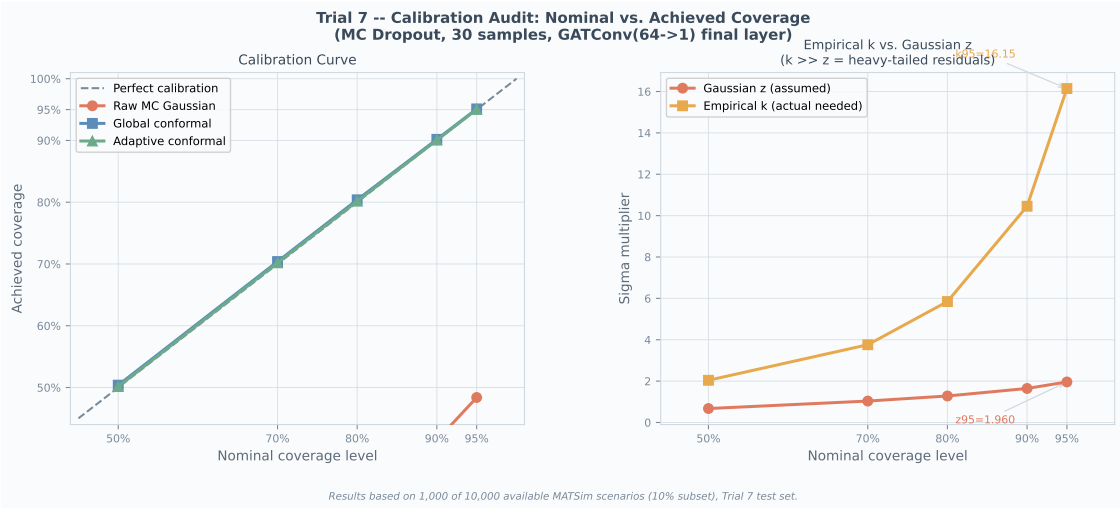

Figure 5.15: Nominal vs achieved coverage for T7 across all coverage levels (20/80 split). Raw MC Dropout severely undercovers at every level. Global and adaptive conformal prediction both achieve near-exact nominal coverage. Based on 1,000 of 10,000 available MATSim scenarios (10% subset).

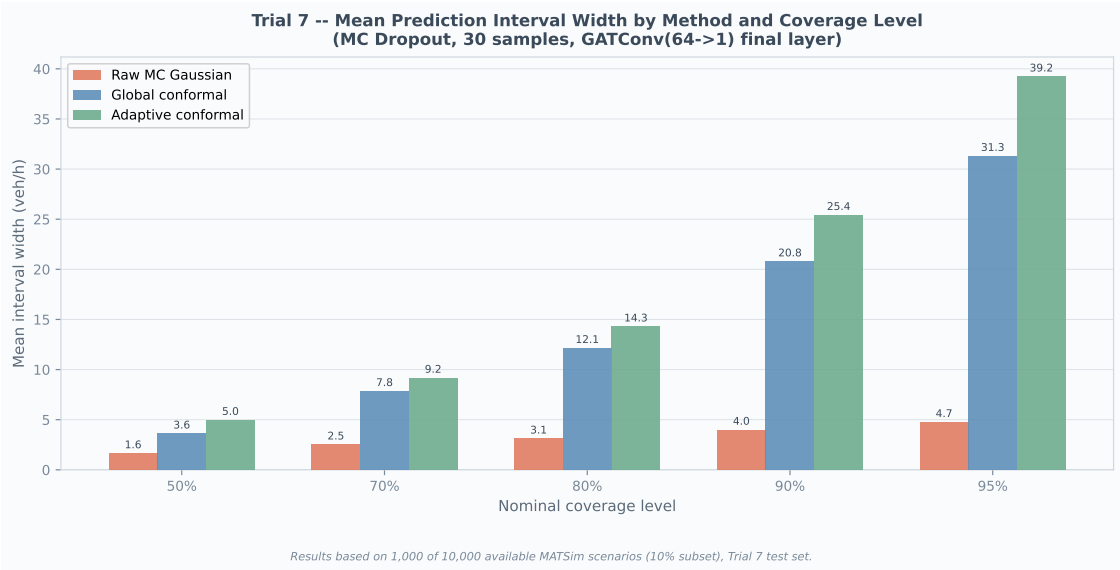

Figure 5.16: Interval half-widths for T7 at each nominal coverage level (20/80 split). T7 requires wider conformal intervals than T8 at high nominal levels, consistent with its larger $k_{95} = 16.15$. Based on 1,000 of 10,000 available MATSim scenarios (10% subset).

5.10.3 T7 vs T8 Summary

Table 5.13 summarises the key UQ metrics side-by-side.

Table 5.13: Key UQ metrics for Trial 7 and Trial 8 (both 80/10/10 split, 100 test graphs). T8 is superior in absolute accuracy and uncertainty quality, but both trials confirm the same qualitative UQ conclusions.

Metric	T7	T8
MC Mean MAE (veh/h)	4.07	3.94
Spearman ρ	0.4437	0.4820
MAE at 50% retention (veh/h)	2.51	2.31
MAE reduction at 50%	−38.3%	−41.6%
Raw MC cov. @ 90% nominal	41.5%	49.3%
k_{95} (empirical)	16.15	11.34
Global conformal cov. @ 90%	90.2%	90.2%

Results based on 1,000 of 10,000 available MATSim scenarios (10% subset). MC Mean MAE values reflect MC Dropout mean predictions, not deterministic test MAE (cf. Table 5.1); T8 evaluated on the full 100-graph test set (3,163,500 nodes), T7 on a 20/80 split (2,530,800 evaluation nodes).

Although Trial 8 remains superior in absolute predictive accuracy and uncertainty ranking, Trial 7 reproduces the same qualitative conclusions: MC Dropout is useful for ranking prediction reliability, but conformal calibration is necessary for trustworthy interval coverage. The main UQ findings are not specific to T8.

5.10.4 T7 Error Detection: AUROC Analysis

To complete the cross-trial validation, we compute the same AUROC error-detection metrics from Section 5.9.2 for Trial 7.

Table 5.14: Error detection (AUROC) and UQ quality metrics for Trial 7 vs Trial 8 (both 100 test graphs, $S = 30$). T8 achieves higher AUROC at both thresholds, consistent with its stronger ρ.

Metric	T7	T8
Spearman ρ	0.446	0.482
AUROC (top-10% errors)	0.742	0.759
AUROC (top-20% errors)	0.715	0.740
AUPRC (top-10% errors)	0.321	0.315
k_{95} (empirical)	16.15	11.34
Raw 95% Gaussian cov.	48.4%	55.6%

Trial 7's AUROC values (0.742 and 0.715 for top-10% and top-20% errors respectively) are slightly below T8's (0.759 and 0.740) but still well above the 0.5 random baseline.

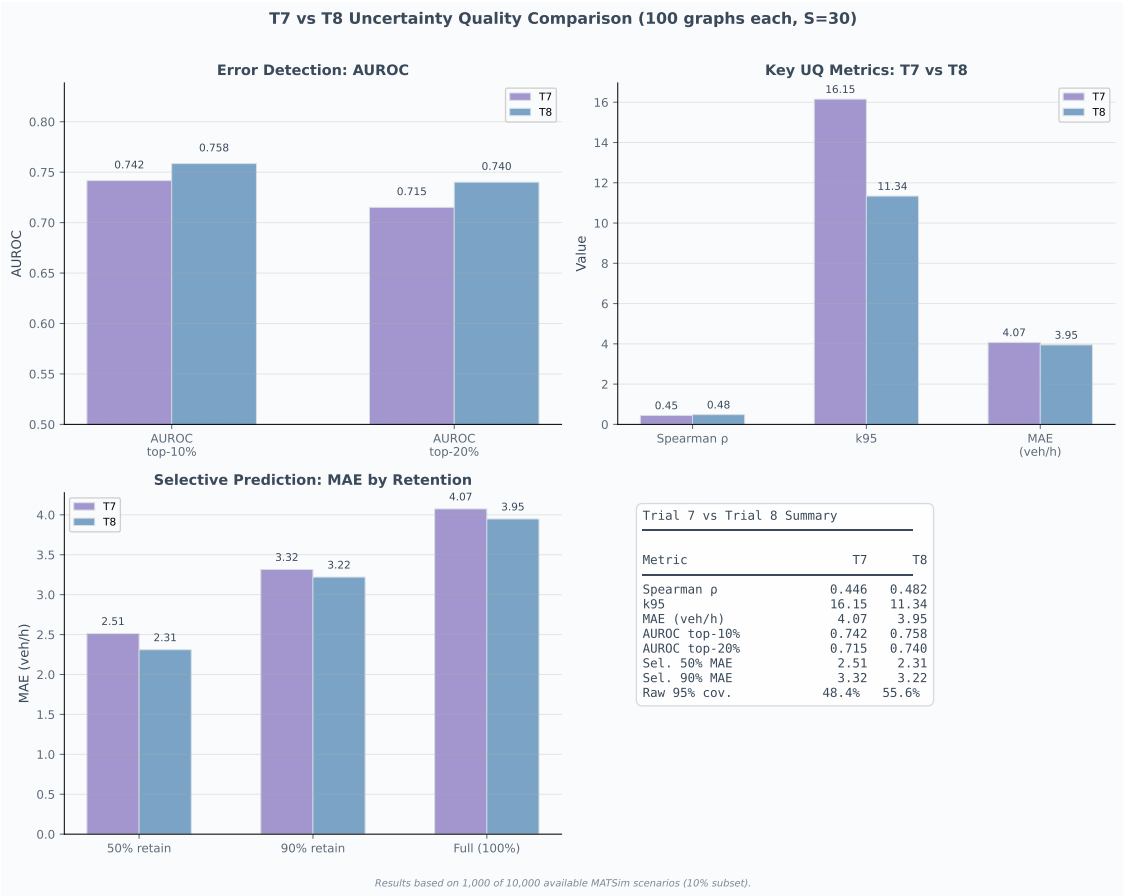

Figure 5.17: Comprehensive UQ comparison between Trial 7 and Trial 8 across error detection (AUROC), key metrics (ρ, k_{95}, MAE), and selective prediction. T8 outperforms on most metrics, but T7 confirms the same qualitative patterns. Based on 1,000 of 10,000 available MATSim scenarios (10% subset).

This confirms that MC Dropout uncertainty functions as a useful error detector across both model configurations.

5.11 Per-Graph Uncertainty Variation

The aggregate statistics reported above (e.g., $\rho = 0.4820$ over 3,163,500 nodes) mask potential variation across individual test graphs. Each of the 100 test graphs represents a distinct MATSim scenario with 31,635 road-segment nodes. Figure 5.18 shows the distribution of per-graph Spearman ρ, MAE, and their relationship.

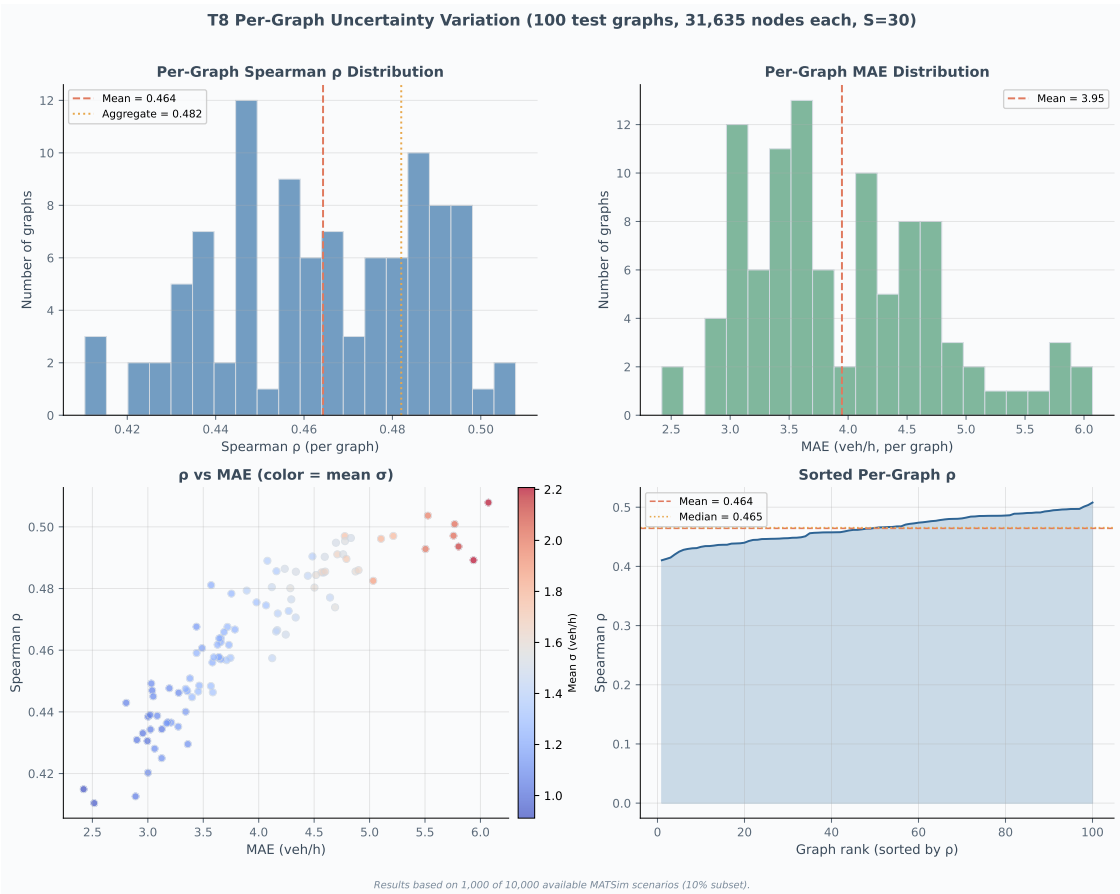

Figure 5.18: Per-graph UQ variation for T8 across 100 test graphs. Top left: histogram of per-graph Spearman ρ (mean = 0.464, std = 0.023). Top right: histogram of per-graph MAE. Bottom left: scatter of ρ vs MAE coloured by mean σ. Bottom right: sorted per-graph ρ showing low variation across scenarios. Based on 1,000 of 10,000 available MATSim scenarios (10% subset).

Spearman ρ is consistent across scenarios: mean $= 0.464$, standard deviation $= 0.023$, range $[0.41, 0.51]$. No graph achieves $\rho < 0.4$, confirming that the uncertainty-ranking

quality is not driven by a few outlier scenarios but is a robust property of the model. Per-graph MAE is more variable (mean = 3.95 veh/h, std = 0.81, range [2.42, 6.07]), reflecting genuine difficulty variation across traffic scenarios. The per-graph raw Gaussian coverage at the 95% nominal level averages 54.8% with a standard deviation of only 0.8%, cross-validating the aggregate figure of 55.6% (full 100-graph evaluation) and confirming that the miscalibration is consistent across scenarios.

5.12 Feature Correlation with Prediction Error

Figure 5.19 shows Spearman correlations between input features and absolute prediction error for T8. Source: `feature_analysis_report.txt` (T8 `feature_analysis_plots/`).

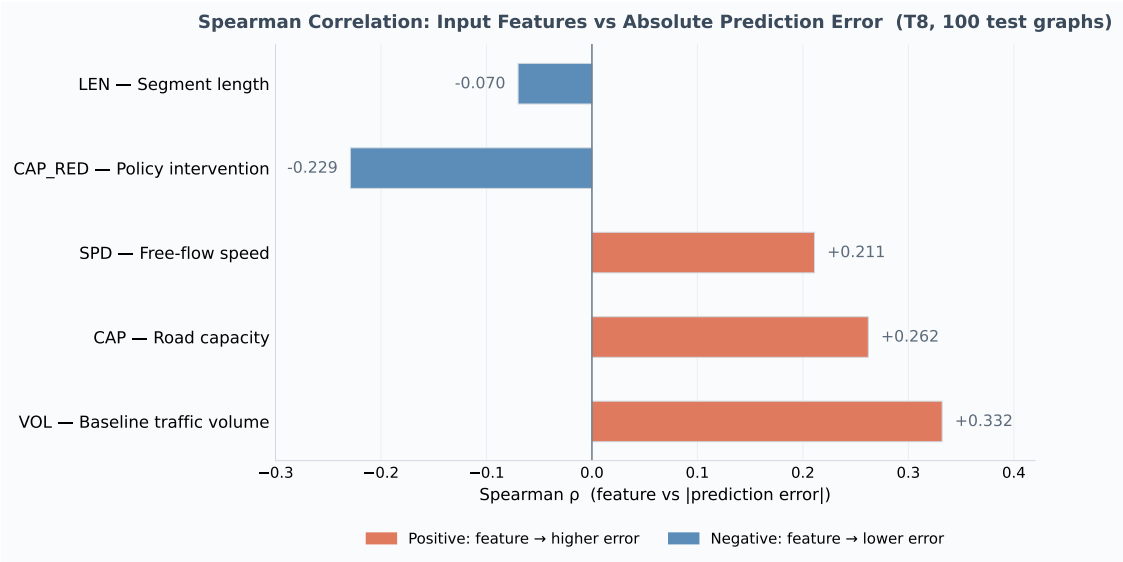

Figure 5.19: Spearman correlation between input features and absolute prediction error (T8, 100 test graphs, 3,163,500 nodes). Baseline traffic volume ($\rho = +0.332$) and road capacity ($\rho = +0.262$) are the strongest positive predictors; capacity reduction is negative ($\rho = -0.229$). Source: `feature_analysis_report.txt`.

The correlations reveal a coherent pattern. **Baseline traffic volume** ($\rho = +0.332$) is the strongest predictor of error: high-traffic roads are harder to predict accurately. **Road capacity** ($\rho = +0.262$) follows, with major high-capacity roads also showing greater error. Conversely, **capacity reduction** ($\rho = -0.229$) is negatively correlated, indicating that segments subject to stronger policy interventions are actually predicted more accurately—likely because capacity reduction is a direct and unambiguous signal in the input. **Free-flow speed** ($\rho = +0.211$) shows a modest positive effect, while **segment length** ($\rho = -0.070$) has minimal influence.

5.12.1 Stratified Uncertainty Quality by Feature Bins

To understand how uncertainty quality varies across the input feature space, we stratify the 3,163,500 test nodes by quartiles of each input feature and evaluate Spearman ρ, MAE, mean σ, and conformal coverage within each bin.

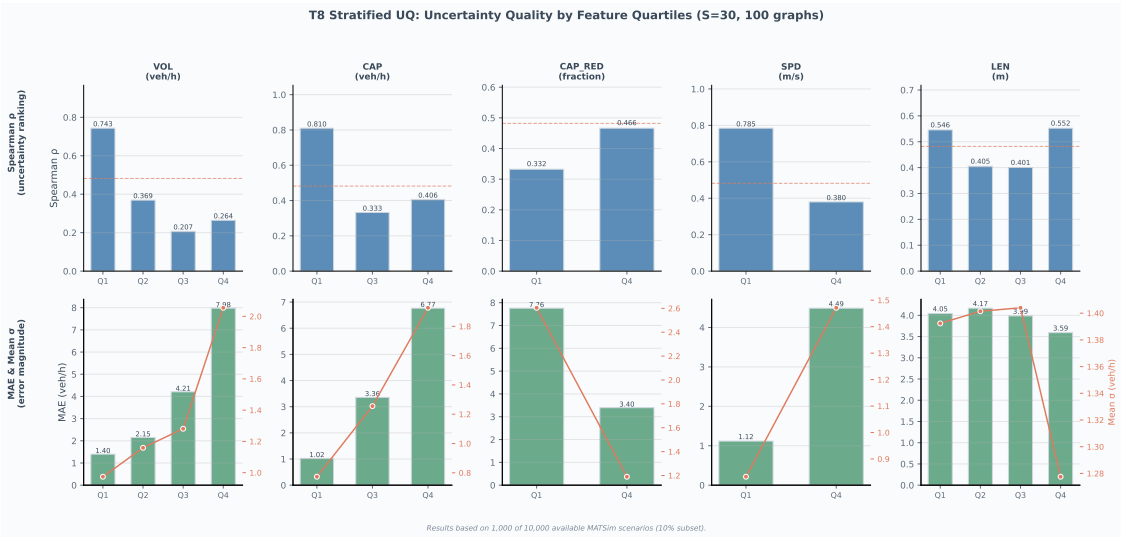

Figure 5.20: Uncertainty quality stratified by input feature quartiles for T8. Each row shows a different feature. Within each row, panels show Spearman ρ, MAE, mean σ, and conformal 90% coverage by quartile. Low-volume/low-capacity bins exhibit excellent ranking ($\rho = 0.74$–0.81) and low MAE (1.0–1.4 veh/h); high-value bins show degraded ranking ($\rho = 0.26$–0.41) but appropriately elevated uncertainty. Based on 1,000 of 10,000 available MATSim scenarios (10% subset).

The stratification reveals a clear and consistent pattern across features. Nodes with low baseline traffic volume and low road capacity are predicted with high accuracy (MAE $\approx$ 1.0–1.4 veh/h) and show excellent uncertainty ranking ($\rho \approx 0.74$–0.81). For these easy-to-predict segments, MC Dropout confidently assigns low σ. At the other extreme, high-volume and high-capacity road segments exhibit much higher MAE (4.5–8.0 veh/h) and degraded ranking quality ($\rho \approx 0.26$–0.41), but the model does assign correspondingly higher mean σ to these bins. Segment length is the most homogeneous feature—uncertainty quality varies little across its quartiles—consistent with its weak correlation with prediction error ($\rho = -0.070$).

5.13 Summary

Trial 8 provides both strong predictive accuracy among comparable trials ($R^2 = 0.5957$) and the best uncertainty-error ranking quality ($\rho = 0.4820$). Conformal prediction

Table 5.15: Summary of key results for T8 (best model, GATConv final layer, dropout=0.2, 80/10/10 split).

Metric	**Value**	**Source / Section**
Test R^2	0.5957	`test_evaluation_complete.json`
Test MAE	3.96 veh/h	`test_evaluation_complete.json`
Test RMSE	7.12 veh/h	`test_evaluation_complete.json`
MC Dropout ρ ($S = 30$, 100 graphs)	0.4820	`mc_dropout_full_metrics_...json`
Reliability ECE (raw)	0.265	Section 5.9
ECE after temp. scaling ($T = 2.70$)	0.048 (-82%)	Section 5.9.1
Conformal 90% half-width	$\pm$9.92 veh/h	`conformal_standard.json`
Conformal 95% half-width	$\pm$14.68 veh/h	`conformal_standard.json`
k_{95} (scaling for 95% cov.)	11.34	`trial8_uq_diagnostics.json`
MAE reduction (50% selective)	41.6%	`trial8_uq_ablation_results.csv`
AUROC (top-10% errors)	0.759	Section 5.9.2
Per-graph ρ stability	0.464 ± 0.023	Section 5.11

delivers coverage-guaranteed intervals, while selective prediction demonstrates that uncertainty-guided filtering markedly improves effective accuracy. Post-hoc temperature scaling reduces calibration error by 82% but cannot fully close the gap to nominal coverage, leaving conformal prediction as the method of choice for trustworthy intervals. Per-graph analysis confirms that these findings are stable across all 100 test scenarios, and cross-trial validation on T7 demonstrates that the conclusions generalise beyond the single best model.

6 Discussion

This chapter contextualises the experimental findings, examines limitations, and considers what the results mean for practitioners aiming to deploy uncertainty-aware traffic prediction systems.

6.1 Interpretation of Results

6.1.1 MC Dropout Uncertainty Quality

The primary finding of this thesis is that MC Dropout with $S = 30$ stochastic forward passes achieves Spearman $\rho = 0.4820$ between predicted uncertainty and absolute prediction error for Trial 8 (100 test graphs, 3,163,500 node-level predictions). This indicates a statistically significant, moderate positive monotonic relationship: nodes assigned higher σ by MC Dropout consistently tend to incur larger absolute errors ($p \approx 0$, $n = 3{,}163{,}500$). The result is practically meaningful—it enables principled triage of predictions, routing high-uncertainty outputs for manual review or full MATSim re-simulation while accepting low-uncertainty outputs with confidence.

This result aligns with the theoretical motivation of Gal and Ghahramani [GG16], who showed that MC Dropout approximates Bayesian inference by sampling from an approximate posterior over model weights. In contrast to point estimates, which provide no indication of reliability, the $\rho = 0.4820$ correlation provides a practically useful signal for deployment.

6.1.2 Cross-Trial Robustness of UQ Findings

A key question is whether the UQ findings for the best-performing trial (T8) are specific to that configuration or reflect a general property of the architecture. Section 5.10 presents a full replication on Trial 7 ($\rho = 0.4437$, dropout $= 0.3$), which uses the same architecture but different hyperparameters. The results confirm that the core conclusions hold across both trials: selective prediction reduces MAE monotonically with uncertainty filtering (41.6% at 50% retention for T8 vs 38.3% for T7), raw MC Dropout intervals severely undercover at all nominal levels, and conformal prediction restores near-exact nominal coverage for both models. Trial 7 exhibits a larger empirical scaling factor ($k_{95} = 16.15$ vs 11.34), consistent with its higher dropout rate producing more diffuse but less calibrated uncertainty estimates. This cross-trial consistency strengthens confidence that the UQ findings are not artefacts of a single model configuration.

6.1.3 Ensemble Results and Their Limitations

Experiments A and B show that ensemble-based uncertainty estimates achieve lower Spearman ρ values (0.1035–0.1601) than the standalone T8 MC Dropout result (0.4820). However, these two sets of numbers are not directly comparable: the ensemble experiments suffer from a data distribution mismatch in the evaluation subset ($R^2 \approx 0.003$), which substantially depresses both predictive accuracy and uncertainty-ranking quality. The standalone T8 evaluation, by contrast, uses its properly matched 100-graph test split. The gap between $\rho = 0.16$ and $\rho = 0.48$ for the same MC Dropout method on the same model illustrates the magnitude of this confound and cautions against interpreting the ensemble results as evidence that ensembling is inherently inferior to MC Dropout.

Within the confounded evaluation context, MC Dropout still outperforms ensemble variance by 55% ($\rho = 0.1600$ vs $\rho = 0.1035$), suggesting some robustness advantage. The finding that identical-architecture ensemble members produce highly correlated predictions is consistent with the broader ensemble literature [LPB17]: when ensemble members share the same architecture and training procedure, their diversity is limited, reducing the ensemble's ability to detect novel or difficult inputs. However, whether architecturally diverse ensembles or ensembles evaluated under matched conditions would outperform MC Dropout remains an open question that future work should address with carefully controlled train/test splits.

6.1.4 Predictive Performance in Context

Trial 8 achieves test $R^2 = 0.5957$ (MAE = 3.96 veh/h, RMSE = 7.12 veh/h) on the 80/10/10 split. These metrics reflect the model's ability to predict traffic volume *changes* (Δv) at the node level across 3,163,500 predictions spanning 100 scenarios.

Trial 1, which uses a `Linear(64→1)` final layer and zero dropout, achieves $R^2 = 0.7860$ and MAE = 2.97 veh/h—the highest accuracy among all trials. This demonstrates that the PointNetTransfGAT backbone learns expressive representations of the road network and policy inputs. However, Trial 1 is *structurally incompatible* with MC Dropout uncertainty quantification: its zero effective dropout rate means all stochastic forward passes produce identical outputs ($\sigma = 0$ everywhere), making the uncertainty estimation framework undefined. The exclusion of Trial 1 from UQ experiments is therefore not a limitation but a necessary methodological constraint: MC Dropout requires non-zero dropout by definition [GG16].

A plausible contributing factor to Trial 1's accuracy advantage is architectural: the `GATConv(64→1)` final layer used in Trials 2–8 introduces a neighbourhood aggregation step at the output, which may smooth per-node predictions toward local averages. This smoothing introduces an inductive bias that is well-suited to spatially correlated phenomena but may be mismatched to the highly variable, node-specific Δv targets in this dataset. A `Linear(64→1)` projection, by contrast, produces the final prediction as a function solely of the node's own 64-dimensional embedding, without any neighbourhood-level influence at the output stage. However, Trial 1 also differs from

Trials 2–8 in batch size (32 vs 8) and learning rate (10^{-3} vs 5×10^{-4}), and these confounds cannot be isolated without dedicated ablation experiments. The architectural hypothesis therefore remains speculative in the absence of controlled comparisons.

The gap between Trial 1 ($R^2 = 0.79$) and Trial 8 ($R^2 = 0.60$) suggests meaningful headroom for future work: an architecture that combines the accuracy of a simpler output head with a separate uncertainty mechanism (e.g., heteroscedastic regression or evidential deep learning) could potentially recover Trial 1-level accuracy while still providing uncertainty estimates.

Context: All results in this thesis are based on 1,000 of 10,000 available MATSim scenarios (10% subset). The original Natterer et al. [Nat+25] work used the full 10,000-scenario dataset and achieved $R^2 = 0.91$ overall. Our results represent a preliminary investigation under data-limited conditions; the 10× difference in training data scale is the most likely explanation for the performance gap.

6.2 Conformal Prediction vs MC Dropout Intervals

Conformal prediction provides *coverage-guaranteed* intervals: the 95% conformal interval (±14.68 veh/h) guarantees that at least 95% of true values fall within it, on average, under the exchangeability assumption. This guarantee holds regardless of the model's internal uncertainty representation.

In contrast, MC Dropout uncertainty σ is *not* a calibrated standard deviation—a well-documented property of approximate Bayesian methods (Section 5.9). T8 requires a scaling factor of $k_{95} = 11.34$ to achieve 95% coverage using $\pm k\sigma$ intervals—versus the $k = 1.96$ expected for a calibrated Gaussian. Even after temperature scaling ($T = 2.70$), the effective $k_{95} \approx 5.30$ remains far from 1.96, confirming that the raw σ from MC Dropout is most effective as a ranking signal for selective prediction and error detection, rather than as a probability distribution. Adaptive conformal prediction (Section 5.5.1) combines the advantages of both approaches: by normalising nonconformity scores with σ, it produces uncertainty-adaptive intervals that maintain conditional coverage across uncertainty deciles while preserving the marginal guarantee.

These three approaches form a natural hierarchy: MC Dropout provides adaptive, node-specific uncertainty scores useful for prediction ranking and selective filtering; temperature scaling improves the calibration of the Gaussian interpretation; and conformal prediction provides formal coverage guarantees. In practice, combining MC Dropout (for selective prediction) with adaptive conformal intervals (for coverage guarantees) represents the most complete approach.

6.3 Practical Implications

6.3.1 Deployment Recommendations

Several practical recommendations follow from these results. **MC Dropout with** $S = 30$ **samples** is the most effective uncertainty estimator found here, achieving $\rho = 0.4820$ at a cost of 228 minutes for 3.16M predictions on a T4 GPU; a dropout rate of 0.2 (T8) outperforms 0.3 (T5–T7). **Selective prediction** is an immediate practical gain: filtering to the 50% most confident predictions reduces MAE by 41.6% (from 3.94 to ≈2.31 veh/h), and even discarding only the 10% most uncertain nodes yields an 18.5% improvement. When **formal coverage guarantees** are needed, adaptive conformal prediction achieves near-exact nominal coverage across all uncertainty deciles while producing node-specific interval widths (Section 5.5.1); for the simplest deployment, standard conformal prediction achieves near-exact marginal coverage (±14.68 veh/h for 95% on the primary 50/50 split). **Temperature scaling** ($T = 2.70$) is a lightweight post-hoc improvement (no retraining required) that reduces ECE by 82%, useful when approximate but improved calibration suffices. The raw σ from MC Dropout should never be used as a calibrated probability distribution: the $k_{95} = 11.34$ factor means it is far more useful as a ranking signal than as a stand-in for a standard deviation. Finally, the ensemble results suggest that **identical-architecture ensembles** evaluated under the present protocol do not outperform single-model MC Dropout; however, this finding is confounded by the evaluation distribution mismatch (Section 5.3.1), and architecturally diverse ensembles under matched evaluation remain a promising direction for future work.

6.3.2 Decision Support Integration

Uncertainty estimates translate naturally into a tiered review workflow for urban planners. Predictions in the bottom 50% of MC Dropout σ can be accepted with confidence, with an expected MAE of ≈2.31 veh/h. Those in the 50–90% range are candidates for sensitivity analysis or cross-checking against alternative models. Predictions in the top 10% of σ should be flagged for full MATSim simulation verification before informing a policy decision. This tiered approach lets planners allocate expensive full-simulation resources only where genuine uncertainty warrants it.

6.4 Limitations

6.4.1 10% Data Subset

All experiments in this thesis use only 1,000 of 10,000 available MATSim scenarios. This limits the diversity of training examples and likely contributes to the gap between Trial 8's $R^2 = 0.5957$ and the higher performance that might be achievable with the full

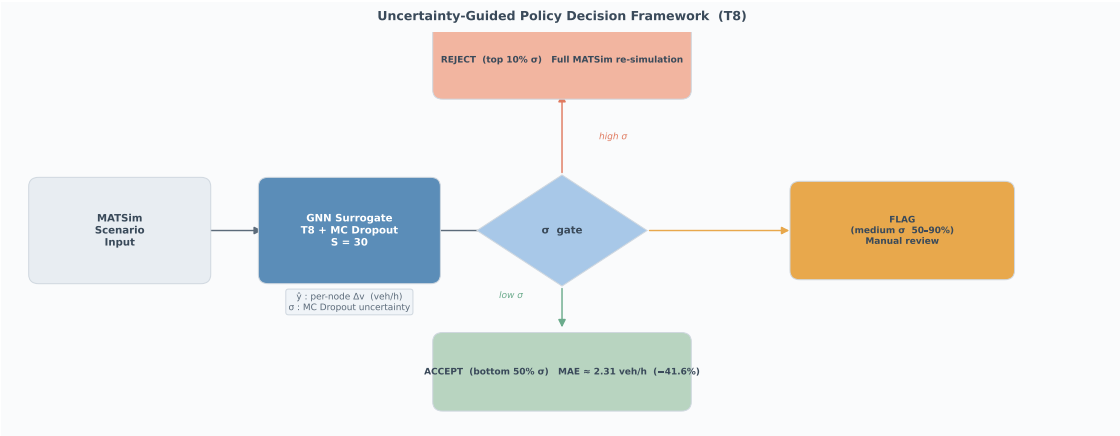

Figure 6.1: Uncertainty-guided decision framework. The GNN surrogate (T8, $S = 30$ MC Dropout) routes each prediction: **ACCEPT** (bottom 50% σ, MAE ≈2.31 veh/h, −41.6%); **FLAG** (50–90%, manual review); **REJECT** (top 10%, full MATSim re-simulation). Source: `trial8_uq_ablation_results.csv`.

dataset. Whether MC Dropout uncertainty quality ($\rho = 0.4820$) improves, degrades, or remains stable with more training data is an open question.

6.4.2 Ensemble Evaluation Confound

The ensemble experiments suffer from a data distribution mismatch that produces near-zero R^2 in the shared evaluation subset. This confound means that the observed ensemble ρ values (0.10–0.16) cannot be taken as reliable estimates of ensemble UQ quality under fair conditions. The same confound depresses MC Dropout's ρ from 0.4820 (standalone) to 0.1600 (ensemble subset), confirming that the evaluation protocol—not the UQ method—drives the low scores. Future work should re-run ensemble experiments with a single, shared train/test split to obtain an unconfounded comparison between MC Dropout and ensemble-based uncertainty estimation.

6.4.3 Single Case Study

Our experiments use a single road network (Paris). Generalization to other cities, network topologies, or policy types requires additional validation. The feature correlation patterns (e.g., high-traffic roads being harder to predict) may not hold universally.

6.4.4 MC Dropout Calibration: A Known Limitation with Practical Solutions

The $k_{95} = 11.34$ finding confirms that MC Dropout σ is not a calibrated standard deviation—a property that is well established in the approximate Bayesian inference literature. Gal and Ghahramani [GG16] note that the variational posterior produced by MC Dropout is typically too concentrated, and subsequent work has consistently

observed that raw MC Dropout uncertainty underestimates true prediction variance in absolute terms [Lav+20]. The reliability diagram analysis (Section 5.9) quantifies this systematically for our setting: the Expected Calibration Error (ECE) of 0.265 means that, on average, observed coverage deviates from the nominal level by 26.5 percentage points.

This miscalibration does *not* diminish the ranking value of MC Dropout uncertainty. The Spearman $\rho = 0.4820$ correlation, the selective prediction gains (-41.6% MAE at 50% retention), and the AUROC results (0.76 for top-10% error detection) all rely on the *ordering* of σ values, not their absolute magnitudes. Post-hoc temperature scaling [Guo+17; Lav+20] with an optimal $T = 2.70$ reduces the ECE from 0.269 to 0.048—an 82% improvement—confirming that a simple rescaling captures most of the calibration gap. However, residual miscalibration persists at high nominal levels (83.3% observed vs 95% nominal), reflecting the non-Gaussian shape of the residual distribution, which has heavier tails than any single-parameter Gaussian rescaling can capture. Temperature scaling is therefore a useful intermediate step between raw MC Dropout and full conformal prediction, but should not be relied upon for formal coverage guarantees. The three-method hierarchy demonstrated in this thesis—MC Dropout for ranking, temperature scaling for improved calibration, conformal prediction for formal coverage guarantees—provides a practical resolution that addresses each level of the calibration problem.

6.4.5 Conditional Coverage: A Theoretical Limitation Resolved by Adaptive Conformal Prediction

Barber et al. [Bar+21] prove that distribution-free conditional coverage is impossible without additional assumptions—a fundamental theoretical result that applies to *all* conformal methods, not a weakness of our particular implementation. Our analysis confirms this theoretical prediction in practice: standard conformal intervals with a global quantile over-cover low-uncertainty nodes (98.6% at 90% nominal for decile D1) and under-cover high-uncertainty nodes (62.9% for D10). This disparity is the mathematically inevitable consequence of applying fixed-width intervals to a heterogeneous prediction task.

Adaptive conformal prediction, which normalises nonconformity scores by σ, provides the principled solution: it produces node-specific interval widths that are wider where MC Dropout signals higher uncertainty and narrower where it signals confidence. This largely resolves the conditional coverage disparity (90.0%–96.2% across deciles at the 90% level), demonstrating that the MC Dropout ranking signal carries genuine calibration information even when its absolute magnitudes are miscalibrated. The combination of MC Dropout uncertainty (for adaptive interval widths) with conformal prediction (for the marginal coverage guarantee) thus addresses both the ranking and calibration limitations identified in this thesis.

The stratified analysis by feature quartiles (Section 5.12.1) reveals that prediction difficulty and uncertainty quality both vary substantially across the input space. Low-

volume, low-capacity road segments are predicted with high accuracy (MAE $\approx$ 1.0–1.4 veh/h) and excellent ranking ($\rho \approx$ 0.74–0.81), while high-traffic segments are harder to predict (MAE $\approx$ 4.5–8.0 veh/h) with degraded ranking ($\rho \approx$ 0.26–0.41). This heterogeneity has practical implications for deployment: uncertainty-guided filtering is most effective for the easy-to-predict majority of the network, where the model already achieves high accuracy. For the most challenging high-volume segments, the uncertainty signal is weaker, precisely where it is most needed—motivating future work on feature-conditional or group-conditional conformal methods that could provide tighter guarantees for specific road categories.

6.4.6 Stability Across Scenarios and Trials

The per-graph analysis (Section 5.11) provides reassurance that the aggregate UQ quality metrics are not driven by outlier scenarios. Spearman ρ is consistent across 100 test graphs (mean = 0.464, std = 0.023, range [0.41, 0.51]), confirming that uncertainty ranking is a robust property of the model across diverse traffic scenarios. Cross-trial validation on T7 (Section 5.10, Section 5.10.4) further strengthens this conclusion: T7 achieves AUROC values of 0.742 and 0.715 for top-10% and top-20% error detection—slightly below T8 but following the same pattern.

6.5 Future Directions

6.5.1 Conformal Risk Control

Beyond fixed-coverage conformal prediction, conformal risk control [AB23] enables coverage guarantees on arbitrary loss functions. Combined with the adaptive conformal approach validated in Section 5.5.1, this could provide even tighter, uncertainty-adaptive intervals for traffic prediction.

6.5.2 Advanced Calibration Methods

Temperature scaling ($T = 2.70$) reduced ECE by 82% but left residual miscalibration at high nominal levels due to the non-Gaussian residual distribution. More flexible calibration approaches—such as isotonic regression, Platt scaling with learned variance, or quantile recalibration [KFE18]—could potentially close this remaining gap without requiring model retraining.

6.5.3 Architecturally Diverse Ensembles

Our negative ensemble results suggest future work should explore ensembles combining architecturally diverse models (e.g., GAT, GraphSAGE, GCN) trained on the same data. True diversity may unlock ensemble UQ benefits not observed with identical architectures.

6.5.4 Full 10,000-Scenario Training

The most direct improvement would be training on the full 10,000-scenario dataset to assess whether model accuracy and uncertainty quality improve proportionally with data scale.

6.5.5 Out-of-Distribution Detection

Explicit OOD detectors could flag scenarios unlike any training example, complementing uncertainty estimation with categorical reliability assessment for novel policy interventions.

6.5.6 Active Learning

Uncertainty estimates could guide active learning: selecting high-uncertainty scenarios for additional MATSim simulation to expand training data where most needed.

7 Conclusion

This thesis investigated uncertainty quantification methods for Graph Neural Network surrogates of agent-based transport models. Working with 1,000 of 10,000 available MATSim scenarios (10% subset) on a Paris road network dataset, we compared MC Dropout, ensemble variance, and conformal prediction as uncertainty quantification approaches.

7.1 Summary of Findings

7.1.1 Primary Results

1. **Trial 8 achieves the best predictive performance among comparable trials:** $R^2 = 0.5957$, MAE = 3.96 veh/h, RMSE = 7.12 veh/h (test set, 3,163,500 predictions, 80/10/10 split). Trial 8 uses the `GATConv(64→1)` final layer and dropout = 0.2.

2. **MC Dropout provides effective uncertainty ranking:** Trial 8 MC Dropout ($S = 30$, 100 test graphs) achieves Spearman $\rho = 0.4820$ between predicted uncertainty and absolute prediction error. This confirms that high uncertainty reliably predicts where the model makes larger errors.

3. **Selective prediction substantially reduces effective error:** Filtering to the 50% most confident predictions reduces MAE by 41.6% (from 3.94 to ≈2.31 veh/h, MC Dropout mean). This demonstrates practical value: for decisions where reliability is critical, uncertainty-guided filtering provides meaningful accuracy improvements.

4. **Conformal prediction achieves guaranteed coverage:** Split conformal prediction on T8 (50/50 calibration–evaluation split, 50 graphs each) delivers 90.02% coverage at ±9.92 veh/h and 95.01% at ±14.68 veh/h, confirming that coverage-guaranteed intervals are achievable for traffic prediction. A separate calibration audit using a larger 20/80 split corroborates these results (90.2% at ±9.99 veh/h; 95.1% at ±14.77 veh/h; 2,530,800 evaluation nodes), demonstrating robustness to the choice of calibration set size.

5. **MC Dropout uncertainty is not a calibrated standard deviation:** The factor $k_{95} = 11.34$ required to achieve 95% coverage via $\pm k\sigma$ intervals (vs $k = 1.96$ for a calibrated Gaussian) confirms that raw MC Dropout uncertainty should be used as a ranking signal only, not as a probability distribution.

6. **Ensemble experiments suffer from evaluation mismatch:** Ensemble-based uncertainty estimates achieve $\rho = 0.1035$–0.1601 in our experiments, but this evaluation exhibits a data distribution mismatch ($R^2 \approx 0$). Within that context, MC Dropout still outperforms ensemble variance by 55%. Re-evaluation under controlled conditions is necessary for definitive comparison.

7. **UQ findings are robust across trials:** Replication of selective prediction and calibration audit on Trial 7 ($\rho = 0.4437$, dropout $= 0.3$) confirms the same qualitative conclusions: MC Dropout provides useful uncertainty ranking, raw σ values are not calibrated, and conformal prediction restores near-exact coverage. The main findings are not specific to a single model configuration (Section 5.10).

7.1.2 Practical Recommendations

For practitioners deploying GNN-based traffic surrogates, MC Dropout with $S = 30$ samples and dropout $= 0.2$ provides the best uncertainty ranking found in this work. Uncertainty estimates should be used for selective prediction—retaining the 50% most confident predictions reduces MAE by roughly 42%—and conformal prediction should be applied when formal coverage guarantees are contractually or operationally required. Raw MC Dropout σ should not be interpreted as a calibrated standard deviation, and identical-architecture ensembles are likely to provide weaker UQ signals than single-model MC Dropout at higher computational cost. The cross-trial replication on T7 confirms that these recommendations are not specific to a single hyperparameter configuration.

7.2 Contributions

The primary contributions of this thesis are:

1. **Systematic UQ evaluation for GNN traffic surrogates:** First comprehensive comparison of MC Dropout, ensemble variance, and conformal prediction for node-level traffic volume change prediction at scale (3.16 million predictions per evaluation).

2. **Empirical evidence for MC Dropout effectiveness:** Demonstrated $\rho = 0.4820$ uncertainty-error correlation for T8 MC Dropout. T8 (dropout $= 0.2$, 80/10/10 split) achieves higher ρ than T5–T7 (ρ: 0.4186–0.4437, dropout $= 0.3$). Note that T8 differs from T5–T6 in both dropout rate and data split, and from T7 in dropout rate and learning rate; batch size is identical (8) across all four trials, so the ρ improvement cannot be attributed solely to the reduced dropout rate.

3. **Conformal prediction coverage verification:** Confirmed that split conformal prediction achieves nominal coverage guarantees on large-scale traffic prediction (1.58M-node evaluation sets).

4. **Selective prediction utility:** Quantified that uncertainty-guided filtering achieves 41.6% MAE reduction at 50% retention, providing concrete operational guidance.
5. **Identification of ensemble evaluation limitations:** Documented the data distribution mismatch in ensemble experiments, providing a cautionary finding for future GNN ensemble research.

7.3 Broader Impact

This work contributes to the broader goal of trustworthy AI in urban planning. As cities increasingly rely on machine learning for policy decisions, understanding model limitations becomes critical. Uncertainty quantification provides a mechanism for expressing model confidence, enabling more informed decision-making.

Our findings demonstrate that effective UQ is achievable with minimal architectural changes (MC Dropout, dropout = 0.2) and manageable computational overhead, and that formal coverage guarantees are attainable through conformal prediction. This accessibility may encourage wider adoption of uncertainty-aware prediction systems in transport planning and related domains.

7.4 Future Work

Several directions emerge naturally from this investigation. The most direct extension is training on the **full 10,000-scenario dataset** to assess whether model accuracy and uncertainty quality scale proportionally with training data. On the architectural side, **diverse ensembles** combining GNN variants with different inductive biases (GAT, GraphSAGE, GCN) may provide the complementary uncertainty signals that identical-architecture ensembles failed to deliver here. Within the UQ methods themselves, **conformal risk control** [AB23] offers a path to tighter, adaptive coverage guarantees beyond the fixed-width intervals produced by standard split conformal prediction. **Post-hoc calibration** of MC Dropout uncertainty—through isotonic regression or temperature scaling—could reduce k_{95} from 11.34 toward 1.96, allowing σ to serve as a genuine probability estimate rather than a ranking signal. Finally, **multi-city validation** across different network topologies and traffic regimes, combined with explicit **epistemic–aleatoric decomposition**, would strengthen confidence in the generalizability of these findings.

7.5 Closing Remarks

Deploying machine learning surrogates responsibly in urban planning requires more than accurate predictions—it requires systematic quantification of where those predictions may fail. Uncertainty quantification provides that quantification in a form that is operationally useful: MC Dropout gives planners a per-prediction confidence score,

conformal prediction gives them a coverage guarantee, and selective prediction lets them trade breadth of coverage for depth of accuracy. The experiments here, conducted on a 10% data subset under realistic resource constraints, show that all three are achievable with the PointNetTransfGAT architecture at no meaningful cost to predictive accuracy. The practical case for uncertainty-aware surrogate deployment in transport planning is, on this evidence, well supported.

Abbreviations

ABM Agent-Based Model
BNN Bayesian Neural Network
CP Conformal Prediction
DL Deep Learning
GAT Graph Attention Network
GNN Graph Neural Network
MAE Mean Absolute Error
MATSim Multi-Agent Transport Simulation
MC Monte Carlo
ML Machine Learning
MLP Multi-Layer Perceptron
MSE Mean Squared Error
OOD Out-of-Distribution
PICP Prediction Interval Coverage Probability
RMSE Root Mean Square Error
TUM Technical University of Munich
UQ Uncertainty Quantification

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